

Do Grassroots Organizations Trade Democracy for Influence? Regulator Selection and the Energy Transition

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- ◉ Rapid industrial change: an increasingly important role
- ◉ *Institutions*: Regulator selection shapes industry outcomes
- ◉ **How does regulator selection affect consumer outcomes when the regulated industry itself is changing?**
- ◉ Relevant to the largest heavily regulated industries: agriculture, railroads, insurance, telecommunications

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Setting: the US electricity sector, 1990–2024

New Mexico, 2020: Ending Elected Regulators

OFFICIAL GENERAL ELECTION BALLOT
STATE OF NEW MEXICO ■ NOVEMBER 3, 2020

CONSTITUTIONAL AMENDMENT 1

PROPOSING TO AMEND THE CONSTITUTION OF NEW MEXICO TO PROVIDE THAT THE PUBLIC REGULATION COMMISSION CONSIST OF THREE MEMBERS APPOINTED BY THE GOVERNOR FROM A LIST OF PROFESSIONALLY QUALIFIED NOMINEES SUBMITTED TO THE GOVERNOR BY A NOMINATING COMMITTEE AS PROVIDED BY LAW AND THAT THE COMMISSION IS REQUIRED TO REGULATE PUBLIC UTILITIES AND MAY BE REQUIRED TO REGULATE OTHER PUBLIC SERVICE COMPANIES.

☐ **FOR**

☐ **AGAINST**

Ballot question text: New Mexico Secretary of State, 2020 General

Election Voter Guide

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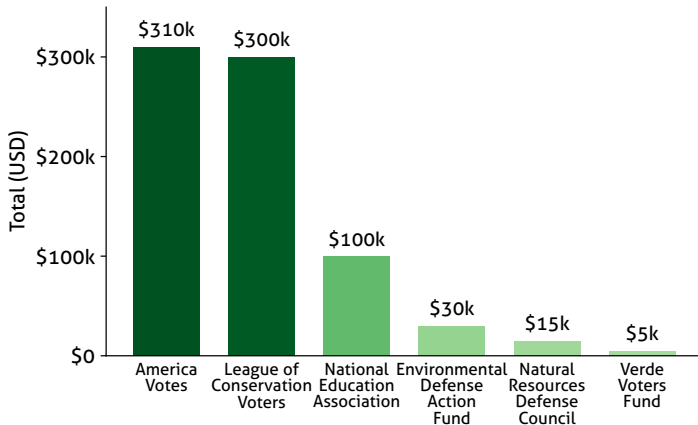
Who campaigns to remove an office from popular election?

The usual suspects? Utilities and fossil fuel interests

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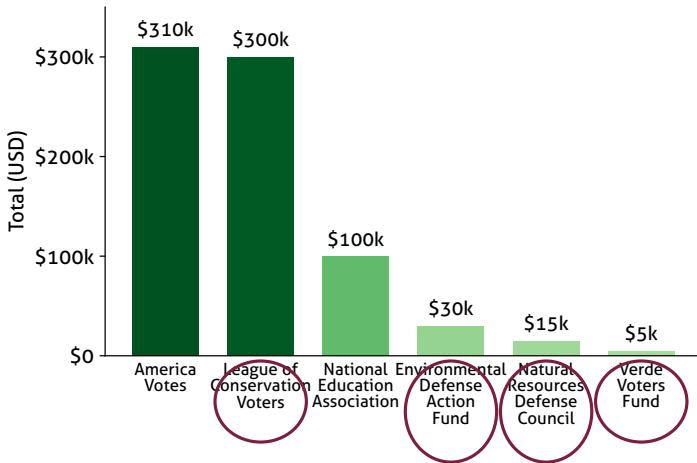
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Who Paid for It?



All group contributions to the campaign for Amendment 1, New Mexico, 2020

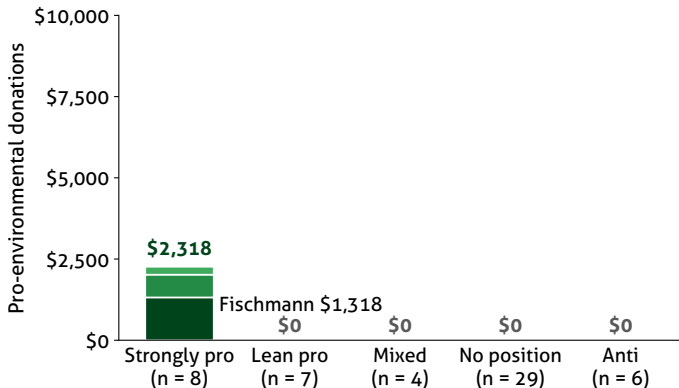
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Not the usual suspects: all six groups are citizen-based nonprofits, and four are climate and environmental groups

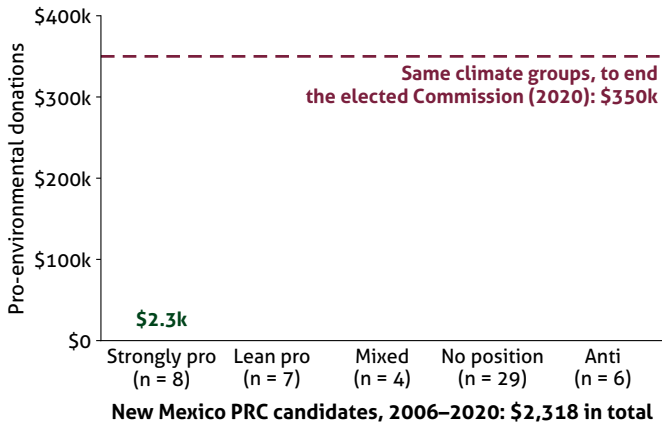
Green Donors Gave 54 PRC Candidates \$2.3k in 15 Years



New Mexico PRC candidates, 2006–2020: \$2,318 in total

54 candidates raised \$3.3M, 0.07% of it from pro-environmental donors.
The Amendment 1 donors gave \$1,500 in total

... and \$350k to End Their Elections



Pro-renewable candidates ran and won, yet the same groups paid \$350k to end Commissioner elections

Follow the Money, 2006–2020; renewable stances hand-coded for every New Mexico PRC candidate.

► Georgia

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Broader lesson: where interest groups seek influence depends on how institutions select decision makers, even for groups built on mass participation

The Standard View: Elections Serve Consumers

Two types of citizens (Besley and Coate 2003)

Consumers (majority)

No stake in the industry → want lower prices

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A simple world: two interests and a single axis of conflict, over price

What Happens When Industries Undergo Change?

Heavily regulated sectors often undergo large-scale industrial change

Industrial change: a new technology or input lets new producers compete with the incumbent industry, and changes when costs and benefits arrive
railroads → trucking landlines → wireless fossil fuels → renewables

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This induces preference heterogeneity *within* existing business and consumer groups: new, cross-cutting coalitions, so regulators no longer face pro-business incentives when appointed and pro-consumer incentives when elected

Cross-Cutting Coalitions: Electricity

	Status quo low price today	Transition invest now, save later
Business	Fossil fuel producers; incumbent utilities	Renewable developers; in-state manufacturers
Consumers	Price-sensitive consumers (the large majority)	Green NGOs; pro-transition citizens (small, organized)

- **Business:** renewable developers need resource plans, rate treatment, and siting rulings that incumbents oppose

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- ◉ **Business:** renewable developers need resource plans, rate treatment, and siting rulings that incumbents oppose
- ◉ **Consumers:** a second type, willing to pay more today: green NGOs, 'grassroots' in membership but staffed by professionals with elite access (Skocpol 2003)

What Is 'Pro-Consumer' Under Industrial Change?

In the standard view, pro-consumer means **low prices today**. Within the energy transition:

- ⊙ New sources need investment now, passed to ratepayers in short term, even where renewables are cheaper

→ What is good for consumers immediately may differ from long run. Most discount the future so prefer the low price today

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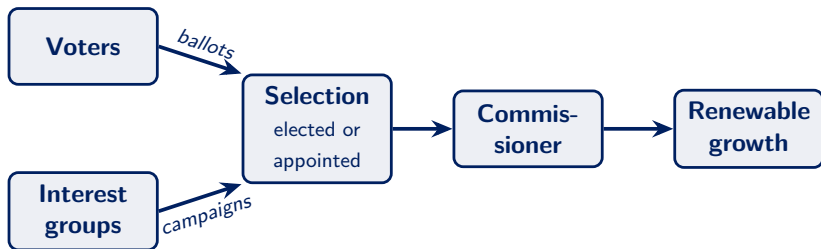
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- ⊙ Regulatory consequences unfold over horizons far longer than electoral cycles

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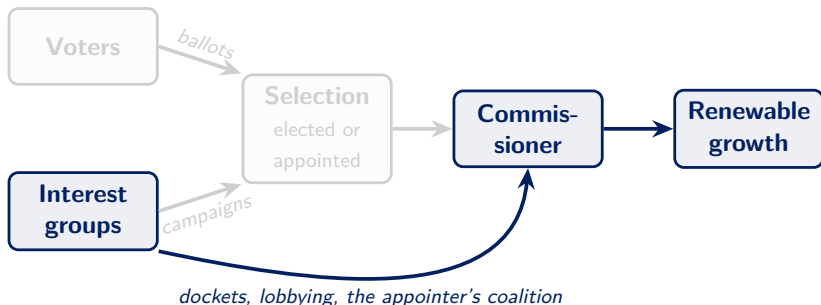
The Argument



Electoral path. Elected Regulators objective function: appease voters, usually based on bills; incumbent industry interest very well organized and dominate campaign settings

► Electoral arena

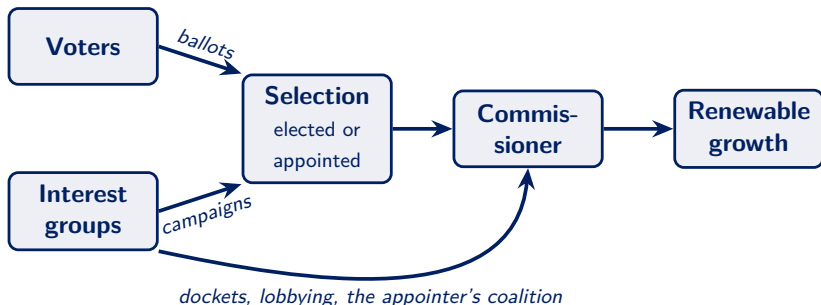
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Regulatory path. Insulated Regulators: can rely on expertise, testimony, regulatory lobbying. Green groups are weak in elections but strong *inside*: elite orgs, persistence in dockets, and often have a seat in the governor's coalition. Appointment gives them a chance to compete

► Inside the beltway

The Argument



⇒ **Renewable growth slower on average when voters elect their Commission**, though partisanship matters.

Setting: Public Utility Commissions (PUCs)

- ◉ Most state PUCs were created in the early 1900s to regulate utilities in the absence of market competition: cost containment, affordable rates

Scope: selection matters where the Commission governs generation (vertically integrated states), less where restructuring or public ownership removed it

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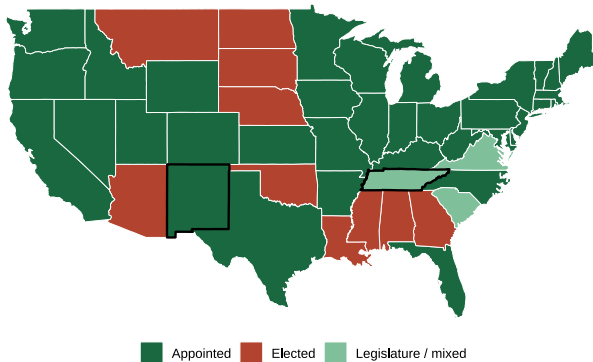
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- ◉ Today far more than price setting: resource plans, rate treatment of new investment, siting and interconnection, reliability
- ◉ Growing demand, cheaper wind and solar, and climate change all push toward integrating alternatives

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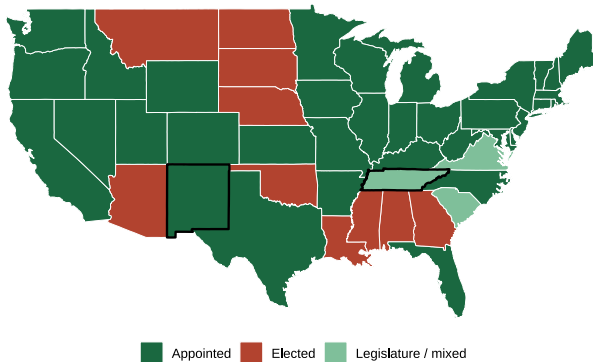
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How States Select Their Commissioners



- About a dozen states elect; the rest appoint (governor, legislature, or mixed)

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- Only two switch, 1990–2024: **TN** (elected → appointed, 1996); **NM** (appointed → elected, 1999; back to appointed, 2023)

Research Design

Exploit state-level variation in institutional design, US, 1990–2024

► Data and sources

1. Selection and renewable generation

Renewable share of generation, 50 states; original roster of every Commissioner and their partisanship; elected vs. appointed Commissions of the same partisan composition, net of geography, economy, and the electorate's partisanship

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3. **Voter behavior**

Every statewide energy ballot measure 2000–2024; every local renewable or climate measure that imposes a direct cost, where complete lists exist; precinct returns; cast vote records linking each vote to the voter's partisan choices (Kuriwaki et al.)

At the same partisan composition, do elected Commissions produce less renewable growth?

The Identifying Variation

- Commission composition varies within nearly every state: 48 of 50 states; within-state SD 0.34 (overall 0.40)

Specification (state s , year t ; institutions and controls lagged one year)

$$\text{Renew}_{st} = \beta_1 \text{Elected}_{s,t-1} + \beta_2 \text{DemComp}_{s,t-1} + \beta_3 \text{Elected}_{s,t-1} \times \text{DemComp}_{s,t-1} \\ + X'_{s,t-1} \gamma + \alpha_{s \text{ or } r} + \delta_t [+ \lambda_s t] + \varepsilon_{st}$$

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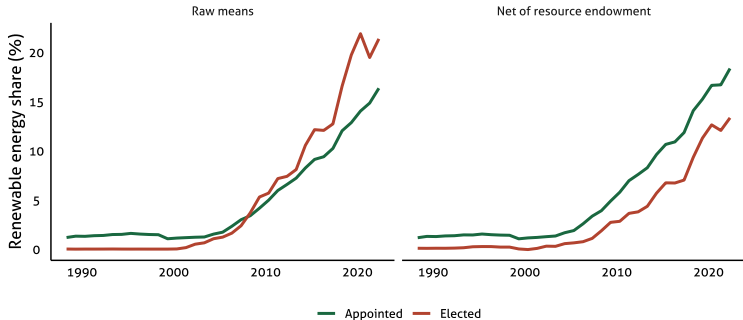
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- ◉ Quantity of interest: elected minus appointed at a 50–50 Commission, $\beta_1 + 0.5 \beta_3$
- ◉ SEs clustered by state, plus inference built for few switching states

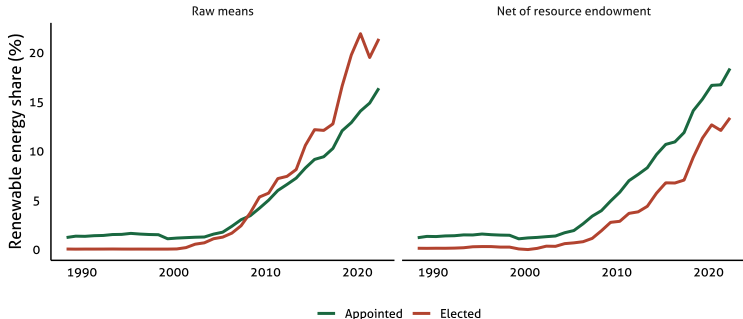
Elected States Sit in the Wind Belt



Mean renewable share by system: raw (left); net of NREL wind and solar endowment within year (right).

- In raw means, elected states generate *more*

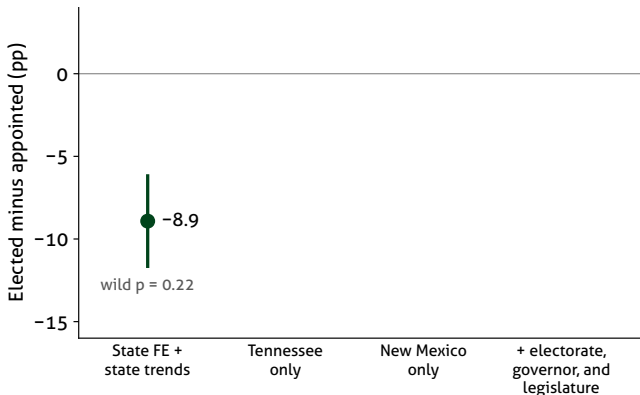
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Mean renewable share by system: raw (left); net of NREL wind and solar endowment within year (right).

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- ◉ Six of twelve elected states are in the Plains wind belt (ND, SD, OK, NE, NM, MT): about double the developable wind of appointed states, which include the low-resource Southeast

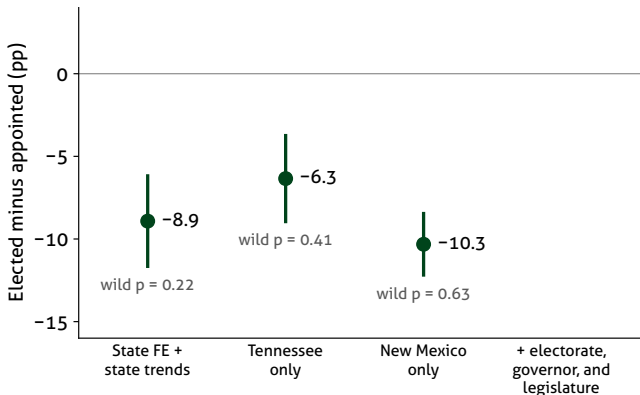
In Tennessee and New Mexico, Renewable Share Was About 9 Points Lower While Elected



State FE and state-specific trends: while elected, renewable share **8.9 pp** lower (at a 50–50 Commission)

State FE absorb each state's endowment; state trends absorb the growing payoff to it (the Plains wind build-out, which coincides with NM's elected period). Two switching states: wild-bootstrap p at right. [► Why two states](#) [► Authority](#) [► Full table](#)

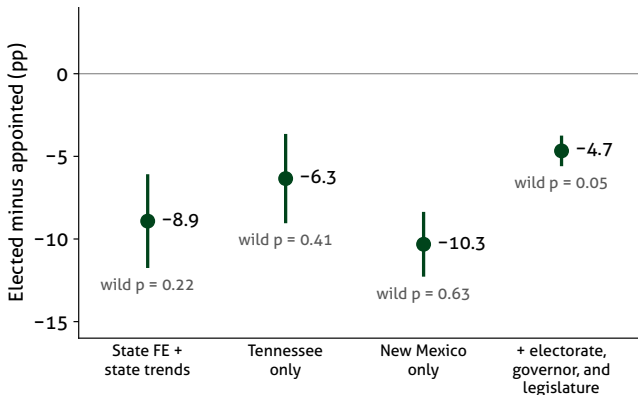
Each Switch Alone Gives the Same Answer



Each switch on its own gives the same answer: from Tennessee's switch alone (New Mexico dropped) **-6.3**; from New Mexico's switches alone (Tennessee dropped) **-10.3**

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Net of State Partisanship, Still About 5 Points Lower



Adding the electorate's partisanship, the governor's party, and the legislature: **-4.7**

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Selection and Generation: What We Learn

Expected: less renewable generation under elected Commissions

Found: in the two states that changed their method, renewable share was about **9 points** lower while elected (about 5 net of state partisanship); each switch alone gives the same sign

Weight: two switching states, so wild-bootstrap $p = 0.22$: two careful case studies, not a precise estimate [► Why two states](#)

→ The mechanism has to carry the argument: **where do green groups compete, and do voters reward the transition?**

Where Green Groups Put Their Effort

Least is known about **pro-renewable groups**: how much they campaign, how much they lobby, and what it buys them

Three things to check: they put more money and time into regulatory work than into campaigns; inside, their resources approach incumbents' while in elections they do not; and for a given effort, their requests fare better before appointed Commissions

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1. **Campaigns**: every contribution in Commissioner races and energy ballot campaigns (Follow the Money)

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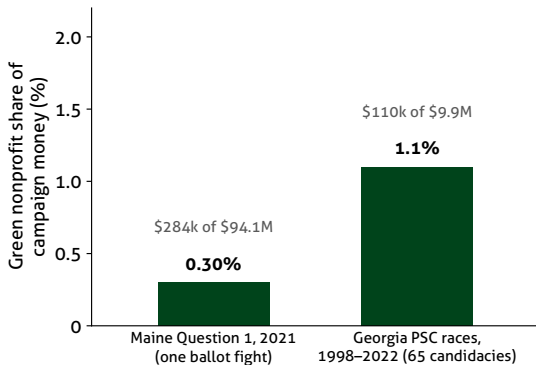
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 - ⊙ Each request's position and its disposition in the final order

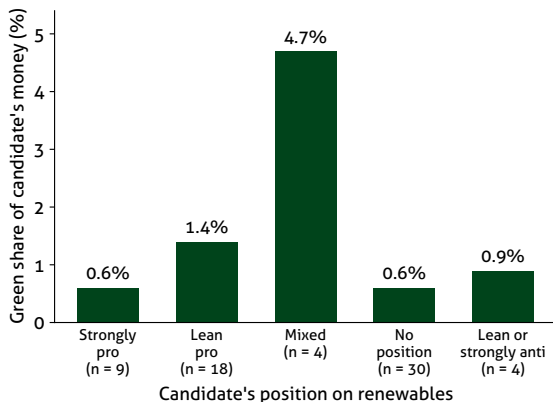
Green Groups Give About 1% of Campaign Money



Green nonprofits supply about 1% or less of campaign money: 0.3% in Maine's ballot fight, 1.1% in 25 years of Georgia PSC races

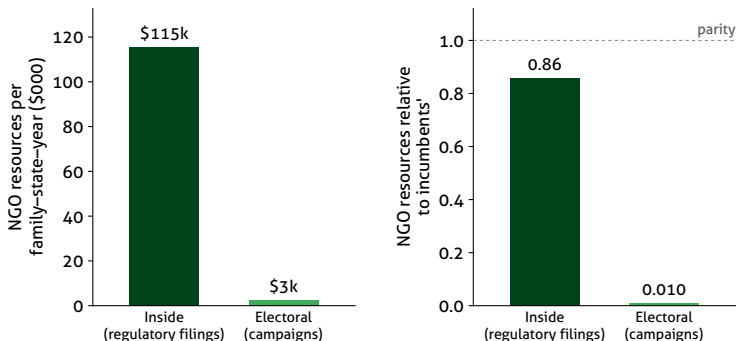
Follow the Money. Maine receipts net of \$13.2M transferred between committees on the same side.

Green Money Not Lacking Because No Green Candidates



Hand-coded renewable positions of all 65 GA PSC candidacies, 1998–2022, linked to every contribution. Sierra Club-endorsed Lindy Miller raised \$1.7M (2018), \$201 of it from pro-environmental donors

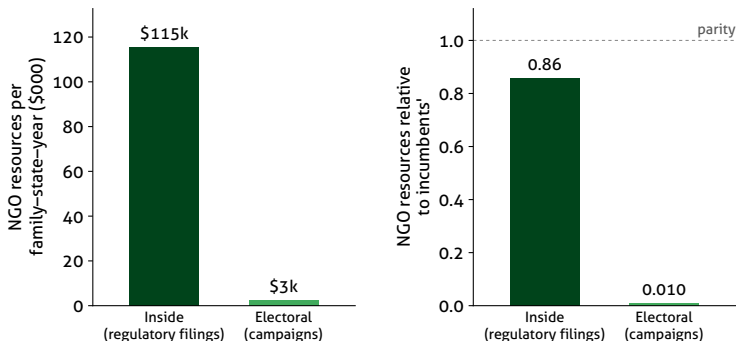
Green Groups Lobby; They Rarely Campaign



Left: mean NGO resources per family-state-year. Right: relative to utility and fossil-firm resources, same sample.

- Same organization, state, and year: \$115k of regulatory work per year vs. \$2.5k of campaigning; 642 hours vs. 20

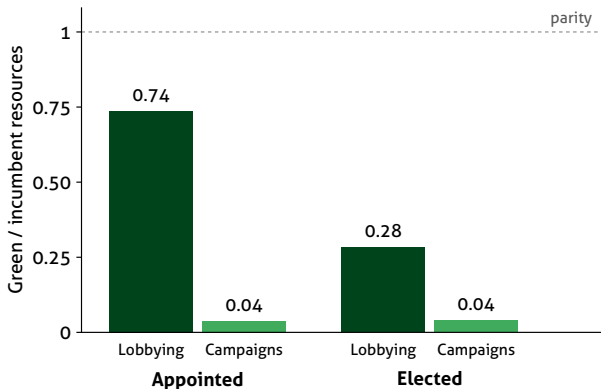
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- Relative to utilities and fossil firms: **0.86** inside vs. **0.01** in elections

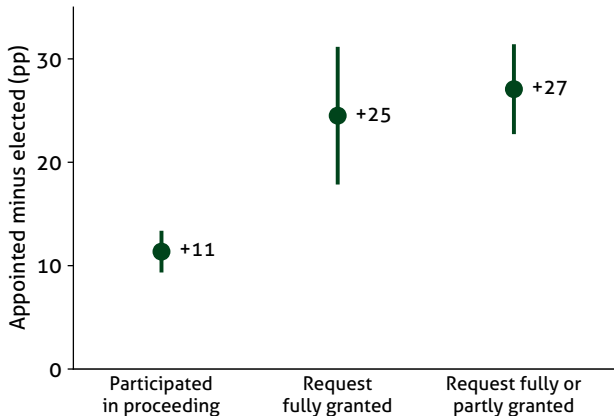
Green Lobbying Rivals Incumbents' Only Under Appointment



Under appointment, green lobbying reaches **74%** of incumbents' resources; under election, **28%**. Campaign resources: 4% in both

[► Tables](#)

Under Appointment, Green Groups Participate More and Win More

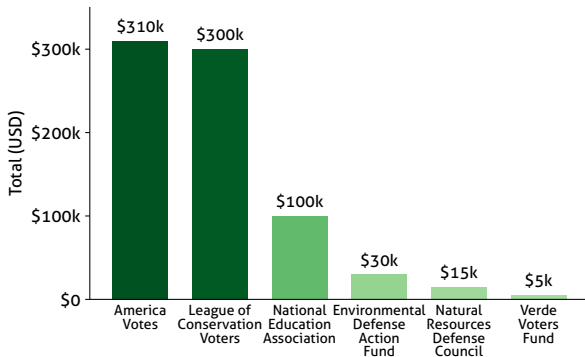


Under appointment, green groups participate more and, for the same effort, win full grants **25 pts** more often

► [Effort and table](#)

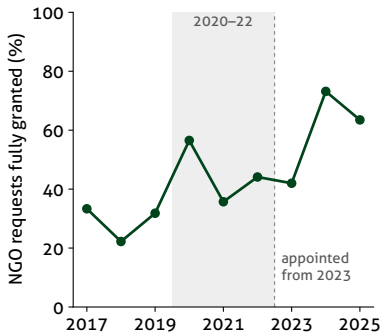
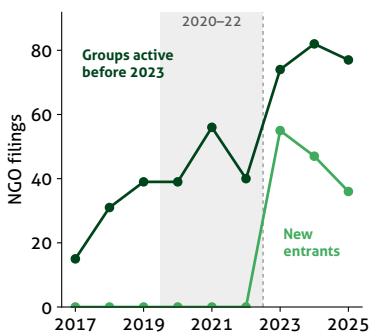
Four matched pairs; 95% intervals, state-clustered. Grants: decided requests, given prior effort.

Green Groups' Biggest Campaign Gift: Ending Commissioner Elections



When green groups do campaign, their largest sums go to ending Commissioner elections. LCV alone gave \$300,000 here: about 85× what the Sierra Club's Georgia chapter gave PSC candidates in 25 years, and about 130× what all pro-environmental donors gave New Mexico's own candidates in 15 years

New Mexico: Before and After Appointment



NM NGO activity and request dispositions. Shading: 2020–2022; dashed line: 2023. Panels have different denominators.

The one state that moved from election to appointment: green filings rise after 2023, both from groups already active and from new entrants, and more requests are granted

► Caveats

Interest Groups: What We Learn

Expected: green groups lobby rather than campaign, and get more for it under appointment

Found

- ⊙ In Commissioner elections, green money is marginal ($\leq 1\%$ of spending) and does not follow pro-renewable candidates

→ Despite their grassroots nature, green groups behave like classic corporate interest groups, and the institution decides how far that gets them

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- ◉ For the same effort, their requests are granted **25 pts** more often before appointed Commissions
- ◉ Revealed preference: offered the chance to abolish Commissioner elections, they paid for it

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Do voters reward the transition at the ballot box?

Testing the Voter Assumption

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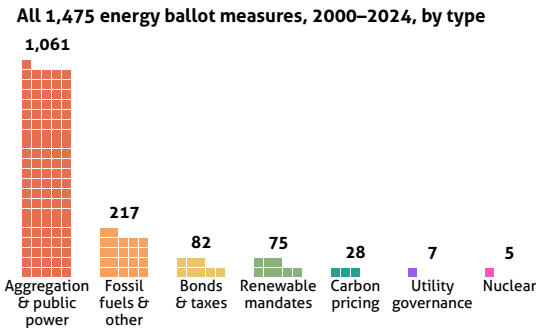
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► Literature

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 - ▶ Literature
- ◉ I measure what voters *do* when renewables carry a cost: statewide and local ballot measures, precinct returns, individual cast vote records

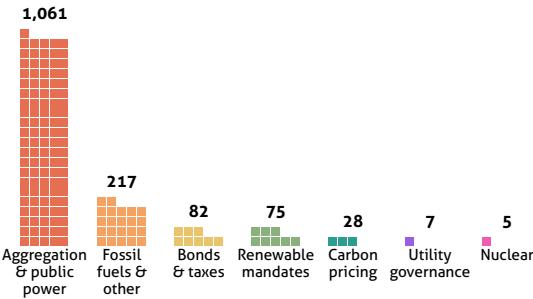
Few Energy Ballot Measures Are About Renewables



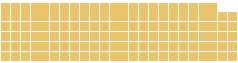
Each square = 10 measures (types with fewer than 10: one square)

Only 26 Statewide Votes on Renewables in 25 Years

All 1,475 energy ballot measures, 2000–2024, by type



Local: 1,422



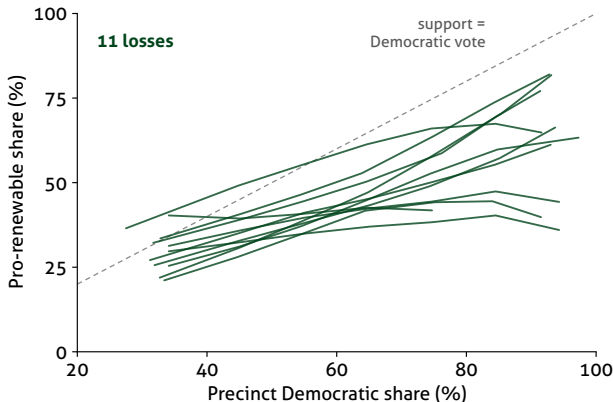
Statewide: 53

teal: the 26 on renewables or climate

Each square = 10 measures (types with fewer than 10: one square)

Pro-renewables won 15 of 26 statewide votes but **lost every carbon-price vote**

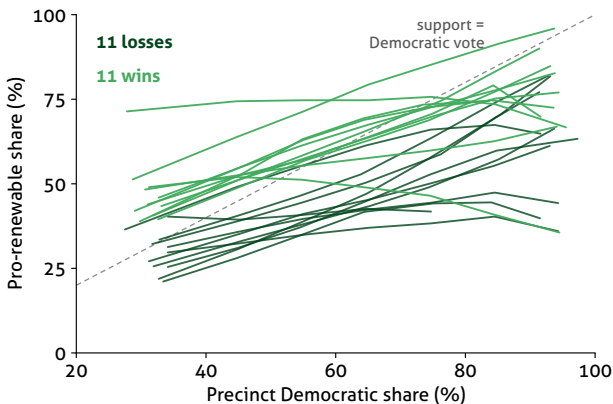
In Deep-Blue Precincts, Losing Measures Ran 27 Points Behind Democrats



All 11 losses: in precincts $\geq 70\%$ Democratic, the measures trailed the Democratic vote by **27 points** (median)

Vote-weighted means in 10-point bins of the Democratic share of the top-of-ticket race; 22 votes with precinct or town returns.

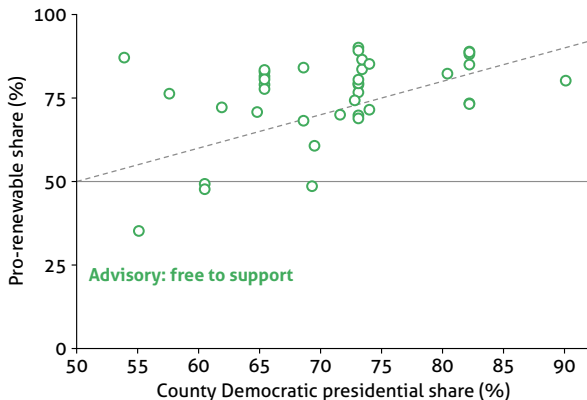
Even Winning Measures Trail the Democratic Vote



The 11 wins run closer to the Democratic vote, but still rise less than one for one [▶ By measure](#)

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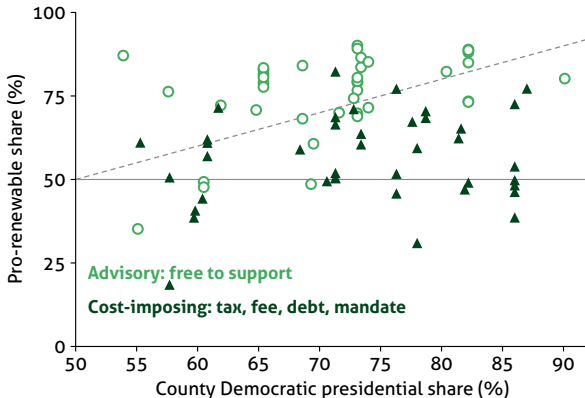
Free Climate Pledges Win Big Locally



Advisory questions (85; 47 with returns): **69%** on average, **80%** where Democrats won $\geq 70\%$

Complete lists where they exist (Ballotpedia; official IL, CA, MA lists), frame fixed before outcomes.

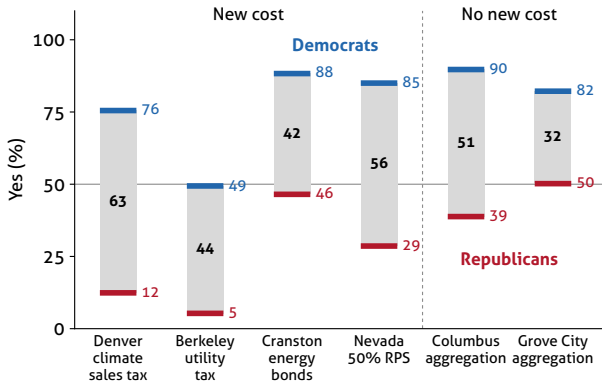
Climate Measures That Cost Something Draw 12 Points Less



Cost-imposing measures (53; 41 with returns): **57%**. Where Democrats won $\geq 70\%$, they lost 10 of 34; within Berkeley and SF, **32 pts** behind advisory [► Full table](#)

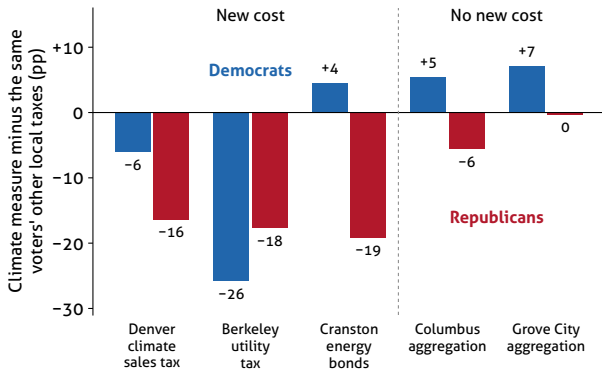
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On Costly Climate Measures, the Parties Split by 42–63 Points



Party gaps of 42–63 pts on the costly measures, vs. **19 pts** on local taxes nationally in 2020

Republicans Give Climate Taxes 16–19 Points Less Than Their Other Taxes



Same voter, same ballot: Republicans discount costly climate measures by **16–19 pts**, by ≤ 6 when there is no new cost; Berkeley's Democrats by 26

► Full

Voters: What We Learn

Expected: little support for renewables once they cost something, even among Democrats

Found: voters support renewables when free, and much less when it costs them

- ◉ **Statewide:** in precincts $\geq 70\%$ Democratic, the 11 losing measures ran a median **27 pts** behind the Democratic vote

→ An elected Commissioner, however Democratic, answers to an electorate that does not reward costly transition policy

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Summary

- ◉ **Outcome:** in the two states that changed how they select regulators, renewables grew more slowly while the Commission was elected

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- ◉ **Civil society:** rational adaptation to US institutions, or a departure from a democratic mandate? Appointment is not a shortcut: appointed Commissions follow the governor [► Scope](#)
- ◉ **Beyond energy:** any policy with immediate costs and delayed, diffuse benefits should face the same electoral headwind

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- ◉ Complete the docket collection for the matched pairs and New Mexico, with a second coder for dispositions
- ◉ Companion paper: the full universe of energy and climate ballot measures since 2000, and more cast vote records

Thank you

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Appendix

- A. Selection and generation
- B. Voters
- C. Interest groups
- D. Anticipated questions
- E. The argument in detail
- F. Moved from the main talk

A. Selection and Generation

Measuring Commission Partisanship

A new roster: 1,287 Commissioner spells, all 50 states, 1990–2026 (name, seat, dates, selecting authority), reconciled against statutory seat schedules

Problem: appointed Commissioners have no official party

- ◉ *Appointer rule* (primary): party of the governor at appointment, or of the selecting legislature or body
- ◉ Staggered terms outlast appointers: a Commission is a layered artifact of past appointments
- ◉ *PUC Democratic Composition*: Democratic share sitting on July 1; 1,848 of 1,850 state-years

Validation: MSU Institute of Public Utilities archive (personal party, 2000–2020)

- ◉ Agreement: 65 of 67 elected (97%); 426 of 595 appointed (72%)
- ◉ Personal-party benchmark (1,464 state-years) used as a check

► Roster

Summary Statistics

	Mean	SD	Min	Max	N
Renewable energy share	0.050	0.090	0.000	0.641	1,750
Renewable and hydro energy share	0.155	0.214	0.000	0.943	1,750
RPS active	0.366	0.482	0.000	1.000	1,750
PUC Democratic composition	0.484	0.403	0.000	1.000	1,748
Elected commission	0.217	0.412	0.000	1.000	1,750
Statewide Democratic share	0.493	0.096	0.218	0.748	1,750
Log population	15.114	1.014	13.025	17.492	1,750
Log GDP	12.093	1.076	9.670	15.012	1,750
Manufacturing share of GDP	0.057	0.016	0.004	0.200	1,750

Notes: State-year panel, 50 states, 1990–2024.

Composition, Selection, and Renewable Share

	Renewable Energy Share				
	(1)	(2)	(3)	(4)	(5)
PUC Democratic composition _{t-1}	-0.002 (0.013)	0.005 (0.011)	-0.000 (0.012)	0.003 (0.011)	-0.006 (0.009)
Elected commission _{t-1}	0.027 (0.032)	0.062 (0.047)	0.043 (0.035)	0.046 (0.035)	-0.103*** (0.021)
Composition × elected _{t-1}	-0.038 (0.050)	-0.028 (0.057)	-0.000 (0.042)	0.003 (0.043)	0.028 (0.019)
RPS active _{t-1}				-0.015 (0.014)	
Comp. effect, elected	-0.040 (0.048)	-0.023 (0.056)	-0.001 (0.040)	0.006 (0.040)	0.021 (0.018)
Observations	1,698	1,698	1,698	1,698	1,698
States	50	50	50	50	50
State fixed effects	No	Yes	Yes	Yes	Yes
Region fixed effects	Yes	No	No	No	No
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	No	Yes	Yes	Yes
State trends	No	No	No	No	Yes

Lagged regressors; HC1 SEs clustered by state; * $p < .10$, ** $p < .05$, *** $p < .01$

◀ Back

Elected–Appointed Differences at Representative Compositions

State and year FE with state-specific trends

PUC Democratic composition	Difference (pp)	Clustered SE	95% CI	p-value
25%	-9.61	1.74	[-13.12, -6.11]	<0.001
50%	-8.92	1.45	[-11.83, -6.01]	<0.001
75%	-8.22	1.27	[-10.78, -5.67]	<0.001
Observations	1,698			
States	50			

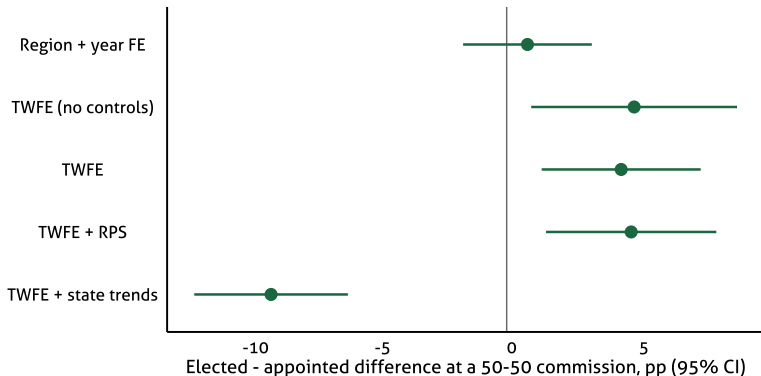
Notes: Elected-minus-appointed renewable share (percentage points) at each lagged commission composition, from the state- and year-fixed-effects model with state trends. State-clustered HC1 standard errors; 49 cluster degrees of freedom.

Inference With Few Switching States (Full)

Specification	Difference (pp)	Cluster SE	p-value			Identifying states	N
			CR1	Wild	Placebo		
Cross-state (region and year FE)	0.8	1.2	0.519	0.557		12	1,698
+ partisanship controls	0.3	0.9	0.741	0.760		11	1,350
Cross-state, Democratic-leaning electorates	-1.5	1.0	0.156	0.143		12	845
State and year FE	4.3	1.5	0.006	0.158	0.142	2	1,698
State FE and state trends	-8.9	1.4	<0.001	0.217	0.009	2	1,698
+ partisanship controls	-4.7	0.5	<0.001	0.054		2	1,350
dropping New Mexico	-6.3	1.4	<0.001	0.407		1	1,664
dropping Tennessee	-10.3	1.0	<0.001	0.632		1	1,664

Notes: The difference is the elected-minus-appointed renewable energy share at a 50–50 commission, in percentage points, from the specification of Table (paper) (one-year-lagged composition, selection system, and economic controls). *Identifying states:* in cross-state rows, the number of states with an elected commission in the sample; in fixed-effects rows, the number of states whose selection system changes, which are the only source of variation in the elected indicator once state fixed effects are included. CR1: standard errors clustered by state, t with $G - 1$ degrees of freedom. Wild: wild cluster restricted bootstrap with Webb weights and 9,999 draws, null imposed. Placebo: Tennessee and New Mexico are dropped, their selection histories are assigned to every ordered pair of the remaining states, and the p -value is the share of placebo estimates at least as large in absolute value as the actual estimate. Democratic-leaning electorates: state-years with a normal-vote Democratic share above 50 percent. Rows adding partisanship control for the normal-vote Democratic share, the governor's party, and the Democratic share of state legislative seats (Snyder state-year file), and therefore cover 1990–2022 and exclude Nebraska.

The Elected–Appointed Difference Across Specifications



Elected minus appointed renewable share at a 50–50 Commission. Whiskers are 95% CIs on cluster df; SEs clustered by state.

Commission Authority and Utility Structure

Every elected Commission is in a vertically integrated state; many appointed ones are restructured

	Retail-choice ctrl		Vert. integrated		PUC covers gen		Both	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
PUC Democratic composition _{t-1}	0.001 (0.011)	-0.006 (0.008)	0.012 (0.015)	-0.001 (0.011)	-0.003 (0.012)	-0.007 (0.009)	0.007 (0.015)	-0.001 (0.012)
Elected commission _{t-1}	0.036 (0.036)	-0.102*** (0.018)	0.027 (0.036)	-0.096*** (0.020)	0.043 (0.033)	-0.117*** (0.016)	0.019 (0.035)	-0.107*** (0.017)
Composition × elected _{t-1}	0.006 (0.042)	0.028 (0.018)	0.013 (0.044)	0.018 (0.021)	0.003 (0.043)	0.026 (0.020)	0.019 (0.044)	0.017 (0.022)
Retail choice active _{t-1}	-0.028** (0.013)	0.022** (0.010)						
Comp. effect, elected	0.008 (0.040)	0.022 (0.017)	0.026 (0.040)	0.017 (0.019)	-0.001 (0.041)	0.019 (0.019)	0.027 (0.040)	0.016 (0.019)
Elect effect at 50%	0.039** (0.016)	-0.088*** (0.012)	0.034** (0.016)	-0.087*** (0.012)	0.044*** (0.014)	-0.104*** (0.010)	0.028* (0.016)	-0.098*** (0.012)
Observations	1,698	1,698	1,190	1,190	1,630	1,630	1,122	1,122
States	50	50	35	35	48	48	33	33
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State trends	No	Yes	No	Yes	No	Yes	No	Yes

Notes: Outcome: renewable share. (1)–(2) add active retail choice; (3)–(4) never-restructured states; (5)–(6) drop Tennessee and Nebraska, whose commissions do not govern generation; (7)–(8) both. Election effect at an evenly split commission. State-clustered HC1 standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Blue States: Composition and Selection Design

	Renewable Energy Share				
	(1)	(2)	(3)	(4)	(5)
PUC Democratic composition $_{t-1}$	-0.014 (0.011)	-0.001 (0.012)	0.003 (0.011)	0.003 (0.011)	-0.004 (0.008)
Elected commission $_{t-1}$	-0.057** (0.024)	-0.026 (0.027)	-0.045* (0.027)	-0.046* (0.027)	-0.111*** (0.021)
Composition $\times$ elected $_{t-1}$	0.084** (0.036)	0.087** (0.040)	0.087*** (0.033)	0.087** (0.036)	0.037 (0.036)
RPS active $_{t-1}$				-0.001 (0.017)	
Marginal effect of composition, elected	0.071** (0.034)	0.086** (0.038)	0.089*** (0.031)	0.090** (0.035)	0.033 (0.035)
Observations	845	845	845	845	845
States	46	46	46	46	46
State fixed effects	No	Yes	Yes	Yes	Yes
Region fixed effects	Yes	No	No	No	No
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	No	Yes	Yes	Yes
State trends	No	No	No	No	Yes

State-years with statewide Democratic share > 0.5. HC1 SEs clustered by state.

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Composition as an Outcome

	Appointer rule (main measure)				Personal party (MSU benchmark)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Elected commission	-0.165** (0.066)	-0.080 (0.057)	0.199*** (0.058)	0.308*** (0.034)	-0.162** (0.066)	-0.076 (0.057)	0.099* (0.059)	0.223*** (0.048)
Voter Democratic share (10 pts)		0.124*** (0.027)	0.016 (0.018)	0.123*** (0.031)		0.082*** (0.028)	0.021 (0.019)	0.104*** (0.026)
Voter share $\times$ elected		0.053 (0.044)	0.152*** (0.038)	-0.015 (0.038)		0.109** (0.046)	0.164*** (0.040)	0.029 (0.037)
Democratic governor			0.607*** (0.044)				0.378*** (0.041)	
Democratic governor $\times$ elected			-0.537*** (0.083)				-0.308*** (0.077)	
Fixed effects	Year	Year	Year	State, year	Year	Year	Year	State, year
State-years	1,748	1,748	1,624	1,748	1,464	1,464	1,393	1,464
States	50	50	50	50	50	50	50	50

Notes: OLS. Outcome: Democratic share of sitting commissioners on July 1. Voter Democratic share (normal vote) is centered at 50% in 10-point units, so *Elected commission* is the elected–appointed difference at an even electorate. Appointer rule: party of the appointing authority; personal party: the commissioner's own. State-clustered standard errors. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

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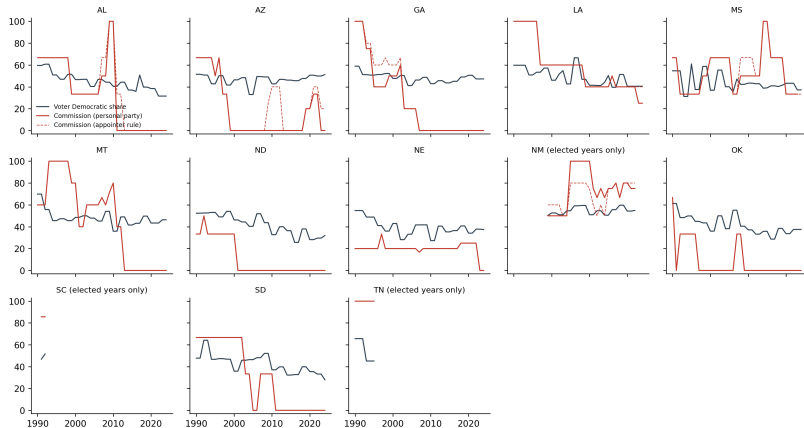
Mismatched Governors and Turnover

455 of 1,262 appointed state-years have a governor whose party runs against the state's lean

System	State lean and governor	State- years	Voter Dem. share	Commission Dem. share	
				Appointer rule	Personal party
Appointed	Democratic-leaning, Democratic governor	442	57.5	82.4	70.6
	Democratic-leaning, Republican governor	272	56.8	21.9	31.1
	Republican-leaning, Democratic governor	183	43.4	85.6	73.1
	Republican-leaning, Republican governor	365	40.9	21.4	31.8
Elected	Democratic-leaning, Democratic governor	53	55.6	67.6	67.6
	Democratic-leaning, Republican governor	58	54.7	51.1	51.4
	Republican-leaning, Democratic governor	51	43.6	29.3	29.0
	Republican-leaning, Republican governor	200	41.3	26.7	25.2
<i>Appointed commissions in mismatch years, by how long the governor's party has held office</i>					
	Democratic-leaning, Republican governor, first two years	66		48.1	48.0
	Democratic-leaning, Republican governor, third year on	206		13.5	25.6
	Republican-leaning, Democratic governor, first two years	45		70.2	64.2
	Republican-leaning, Democratic governor, third year on	138		90.7	75.7

Notes: State-years 1990–2022 with a Democratic or Republican governor. A state leans Democratic when its normal-vote Democratic share is at least 50 percent. Commission shares are means of the state-year Democratic share of party-classified sitting commissioners on July 1; the personal-party benchmark covers fewer state-years. Governor party from the Snyder state-year file.

Composition in the Electing States



Solid red: personal-party measure; dashed red: appointer rule; dark line: voter Democratic share. NM, SC, and TN shown only for elected years.

The RPS Pathway: Stage 1

Controlling for the electorate and legislature, the composition coefficient falls from 0.18 to 0.02

	RPS active (1)	RPS active (2)	RPS active (3)	RPS active (4)	First adoption (5)
PUC Democratic composition _{t-1}	0.183** (0.074)	0.080 (0.065)	0.015 (0.071)	0.115** (0.055)	0.015 (0.023)
Elected commission _{t-1}	-0.216* (0.126)	-0.169 (0.111)	-0.145 (0.106)	0.322*** (0.114)	-0.001 (0.019)
Composition × elected _{t-1}	0.343** (0.168)	0.246* (0.148)	0.218 (0.146)	0.047 (0.127)	0.010 (0.031)
Voter Democratic share		1.627*** (0.250)	0.365 (0.310)	0.214 (0.211)	0.198*** (0.074)
Democratic governor			0.045 (0.042)	0.013 (0.040)	0.009 (0.015)
Democratic legislative seat share			0.841*** (0.159)	1.245*** (0.186)	0.152*** (0.051)
Fixed effects	Region, year	Region, year	Region, year	State, year	Region, year
State-years	1,698	1,698	1,350	1,350	908

Notes: Linear probability models with lagged log population, log real GDP, and manufacturing share. Governor and legislature from the Snyder file (1990–2022; Nebraska missing). Column 5 excludes Iowa and drops states after first adoption. State-clustered standard errors. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

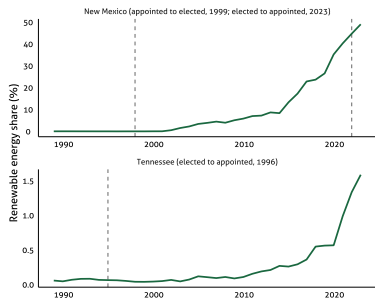
The RPS Pathway: Stage 2

Neither RPS status nor years since adoption predicts the renewable share

	Renewable energy share			
	(1)	(2)	(3)	(4)
RPS active _{<i>t</i>-1}	0.008 (0.016)	-0.013 (0.015)	-0.021 (0.022)	-0.002 (0.009)
Years since RPS adoption _{<i>t</i>-1}			0.001 (0.003)	-0.003 (0.002)
Fixed effects	Region, year	State, year	State, year	State, year, trends
State-years	1,698	1,698	1,698	1,698

Notes: The outcome is the non-hydro renewable share of generation. All columns include one-year-lagged log population, log real GDP, and manufacturing share. Standard errors clustered by state. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Synthetic Control on the Transitions



- TN 1996: -2.5 pp vs. synthetic (rank $p = 0.59$), opposite to the prediction
- NM 1999: $+6.8$ pp ($p = 0.10$), opposite to the prediction
- NM 2023: two post-years, poor pre-fit
- Pooled counterfactual estimator (Liu, Wang, Xu 2024): 0.015 (SE 0.101)

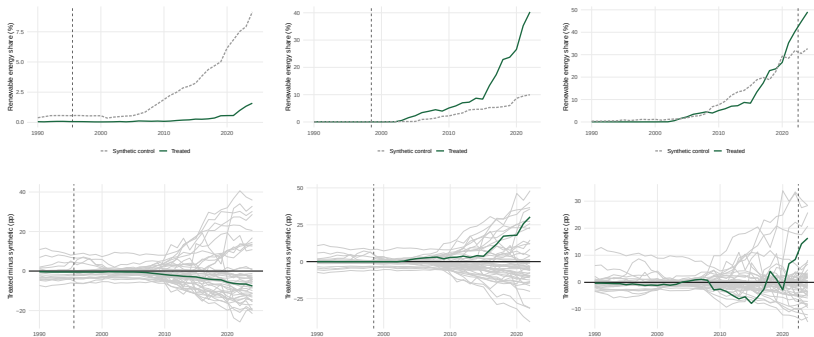
Uninformative rather than disconfirming: non-hydro renewables were near zero everywhere until the mid-2000s, so the 1990s pre-periods are flat lines any donor mix matches (NM pre-RMSPE 0.01 pp); post-1999 NM growth reflects the legislature's RPS and the national wind build-out; TN is supplied by TVA

Synthetic-Control Estimates

	Tennessee (elected to appointed, 1996)	New Mexico (appointed to elected, 1999)	New Mexico (elected to appointed, 2023)
Average post-transition gap (pp)	-2.49	6.81	15.29
Pre-transition RMSPE (pp)	0.45	0.01	3.33
Post/pre RMSPE ratio	7.4	852.9	4.6
Permutation rank p -value	0.587	0.104	0.146
Leave-one-donor-out range (pp)	[-4.32, -1.37]	[3.99, 6.80]	[15.15, 25.14]
Augmented SCM (ridge) gap (pp)	—	—	—
90% conformal interval	[—, —]	[—, —]	[—, —]
Positive-weight donors	39	7	5
Placebo units	45	47	47

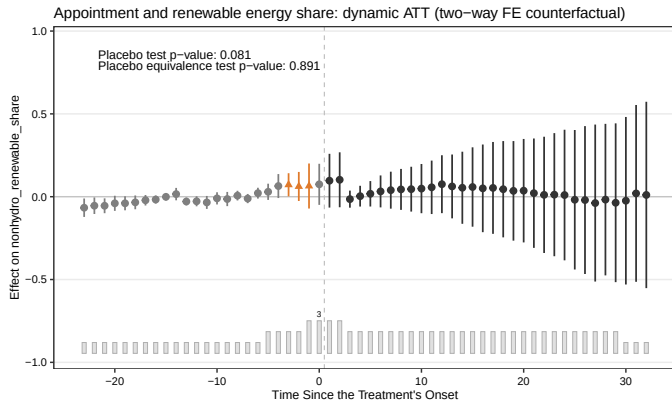
Notes: Outcome: renewable share. Donors: states that never change selection method. Rank p : share of placebos with a post/pre RMSPE ratio at least the treated state's. Augmented: ridge-corrected estimator (Ben-Michael, Feller, and Rothstein 2021) with a conformal interval.

Synthetic Control: Paths and Placebos



Left to right: Tennessee 1996, New Mexico 1999, New Mexico 2023. Top: treated vs. synthetic. Bottom: treated gap (bold) against in-space placebos.

Pooled Counterfactual Estimator



Effect of appointment on the renewable share by time relative to the change, all three transitions, treatment allowed to reverse. 95% bootstrap bands.

Including Conventional Hydro

	Renewable and Hydro Energy Share				
	(1)	(2)	(3)	(4)	(5)
PUC Democratic composition _{<i>t</i>-1}	0.068 (0.052)	0.001 (0.012)	-0.008 (0.013)	-0.007 (0.014)	-0.009 (0.008)
Elected commission _{<i>t</i>-1}	0.059 (0.079)	0.040 (0.029)	0.029 (0.026)	0.030 (0.027)	-0.083*** (0.027)
Composition × elected _{<i>t</i>-1}	-0.122 (0.095)	0.015 (0.038)	0.024 (0.034)	0.025 (0.034)	0.034* (0.019)
RPS active _{<i>t</i>-1}				-0.003 (0.025)	
Comp. effect, elected	-0.054 (0.073)	0.016 (0.033)	0.016 (0.029)	0.018 (0.030)	0.024 (0.018)
Observations	1,698	1,698	1,698	1,698	1,698
States	50	50	50	50	50
State fixed effects	No	Yes	Yes	Yes	Yes
Region fixed effects	Yes	No	No	No	No
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	No	Yes	Yes	Yes
State trends	No	No	No	No	Yes

Outcome includes conventional hydro. HC1 SEs clustered by state.

The Commissioner Roster: Remaining Limitations

- ◉ Historical seat capacity is documented unevenly: 1,516 of 1,850 state-year capacity records are assumed (usually current size carried back); 41 states rely on an assumed capacity for at least one year
- ◉ Matters for the structural audit, not the composition measure, which uses only Commissioners recorded as serving
- ◉ 66 July 1 snapshots exceed assumed capacity; 153 overlapping-service spans remain (holdovers, contested succession dates)
- ◉ 240 state-years below the authorized maximum: vacancies, or maxima above normal staffing (New York)
- ◉ 215 spells carry a manual-review flag, mostly successor-bounded end dates
- ◉ 16 of 1,287 spells unclassified: 6 elected without party evidence (LA, MS, MT); 10 chosen by divided legislative or mixed selectors (SC, TN, VA). Costs 2 state-years of coverage

B. Voters

Energy Ballot Measures by Type, 2000–2024

Measure type	Statewide	Local	Total
Fossil fuels, facility siting, and other energy	13	204	217
Bond, tax, and finance	12	70	82
Renewable mandates and subsidies	11	64	75
Utility governance	7	0	7
Public power, transmission, and aggregation	5	1,056	1,061
Carbon pricing and climate policy	5	23	28
Nuclear	0	5	5
All energy measures	53	1,422	1,475

Notes: Statewide: the complete verified list (53). Local: the local dataset so far (v1.8; 1,422), not a national census; mostly Illinois aggregation referenda (989), California (95), and Arizona franchise votes (67). Types from measure text: carbon pricing includes building electrification; public power includes aggregation; the residual group covers fossil fuels, siting, and other energy.

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Statewide Renewable and Climate Votes, 2000–2024

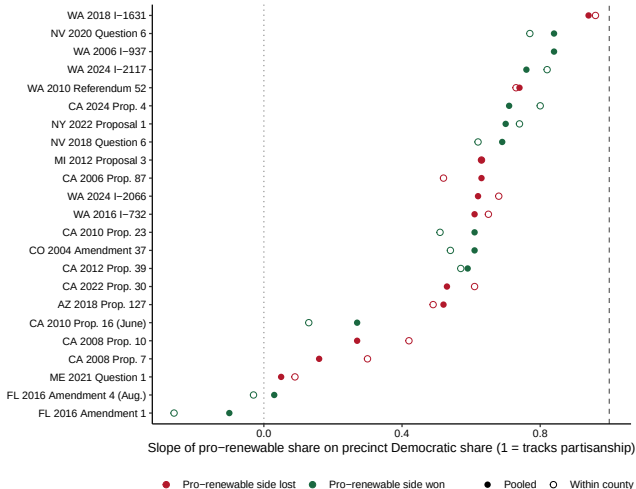
State	Year	Measure and policy	Pro-renewable vote	Outcome for renewables
CO	2004	Amendment 37 — renewable energy requirement	Yes	Win
CA	2006	Proposition 87 — alternative energy funding	Yes	Loss
WA	2006	Initiative 937 — renewable energy standard	Yes	Win
CA	2008	Proposition 7 — renewable energy standards	Yes	Loss
CA	2008	Proposition 10 — alternative fuel and renewables	Yes	Loss
MO	2008	Proposition C — utility RPS mandate	Yes	Win
FL	2008	Amendment 3 — renewable energy devices excluded from property assessment	Yes	Win
ME	2010	Question 2 (primary) — renewable energy bond	Yes	Win
CA	2010	Proposition 16 (primary) — utility-backed supermajority for public power and community choice [†]	No	Win
CA	2010	Proposition 23 — suspend the state climate law (AB 32)	No	Win
WA	2010	Referendum 52 — energy-efficiency bonds for schools	Yes	Loss
MI	2012	Proposal 12-3 — 25% RPS by 2025	Yes	Loss
CA	2012	Proposition 39 — corporate tax change funding clean energy	Yes	Win
FL	2016	Amendment 1 — utility-backed restriction on rooftop solar [†]	No	Win
FL	2016	Amendment 4 (primary) — solar tax exemption	Yes	Win
WA	2016	Initiative 732 — revenue-neutral carbon tax	Yes	Loss
AZ	2018	Proposition 127 — 50% RPS by 2030	Yes	Loss
NV	2018	Question 6 — 50% RPS by 2030	Yes	Win
WA	2018	Initiative 1631 — carbon fee	Yes	Loss
NV	2020	Question 6 — 50% RPS by 2030 (required second vote)	Yes	Win
ME	2021	Question 1 — block NECEC clean-energy corridor [†]	No	Loss
NY	2022	Proposal 1 — environmental bond incl. climate mitigation and clean energy	Yes	Win
CA	2022	Proposition 30 — income tax for zero-emission vehicles and wildfire prevention	Yes	Loss
CA	2024	Proposition 4 — climate bond incl. clean energy	Yes	Win
WA	2024	Initiative 2066 — prohibit restrictions on natural gas [†]	No	Loss
WA	2024	Initiative 2117 — repeal cap-and-invest program	No	Win

Statewide Votes by Precinct Partisan Lean

	Result	Pro-	Dem.	Slope	Dem. share	Pro – Dem.	Strong Dem. (>60%)		Lean Dem. (50–60%)	
		renewable	share		at 50% pro	(Dem. $\geq$ 70%)	Pro majority	Median pro	Pro majority	Median pro
CO 2004 (Amendment 37)	W	53.6	47.6	0.61	41.5	-5.1	84	68	83	57
CA 2006 (Prop. 87)	L	45.4	41.0	0.63	48.5	-13.0	86	62	64	54
WA 2006 (I-937) ^d	W	51.7	58.7	0.84	56.7	–	100	60	53	50
CA 2008 (Prop. 7)	L	35.6	62.3	0.16	>100	-42.6	8	38	3	34
CA 2008 (Prop. 10)	L	40.6	62.3	0.27	97.4	-35.8	34	46	9	40
CA 2010 (Prop. 16, June) ^c	W	54.6	53.3	0.27	36.2	-19.4	61	56	57	52
CA 2010 (Prop. 23)	W	61.5	56.8	0.61	37.9	-5.2	98	72	93	61
WA 2010 (Referendum 52)	L	46.1	52.1	0.74	57.2	-10.9	84	62	27	46
CA 2012 (Prop. 39)	W	61.1	61.9	0.59	42.9	-8.9	99	70	93	57
MI 2012 (Proposal 3)	L	37.7	54.8	0.63	74.0	-28.7	66	56	1	37
FL 2016 (Amendment 1) ^e	W	49.0	49.4	-0.10	–	-40.2	26	44	53	51
FL 2016 (Amendment 4, Aug.) ^c	W	72.6	44.3	0.03	–	-12.2	98	74	98	76
WA 2016 (I-732)	L	40.8	58.8	0.61	73.5	-26.6	48	50	1	38
AZ 2018 (Prop. 127)	L	32.1	52.1	0.52	86.6	-35.5	21	39	0	35
NV 2018 (Question 6)	W	59.3	52.2	0.69	38.6	-3.9	98	71	96	64
WA 2018 (I-1631)	L	43.5	58.5	0.94	65.4	-15.1	67	56	1	39
NV 2020 (Question 6)	W	57.9	51.2	0.84	41.5	-1.7	100	71	97	63
ME 2021 (Question 1) ^d	L	40.8	54.5	0.05	–	-36.5	6	40	6	40
CA 2022 (Prop. 30)	L	42.4	59.2	0.53	73.5	-26.9	47	50	9	41
NY 2022 (Proposal 1) ^b	W	76.6	62.8	0.70	26.1	5.9	100	92	100	72
CA 2024 (Prop. 4)	W	59.8	60.4	0.71	46.7	-6.4	98	69	82	58
WA 2024 (I-2066)	L	48.3	59.5	0.62	62.3	-17.3	72	54	15	45
WA 2024 (I-2117)	W	62.0	59.5	0.76	43.9	-1.7	100	69	96	59

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Slopes on Precinct Partisanship



Slope of pro-renewable share on Democratic share, pooled (filled) and within county (hollow). One = support tracks partisanship exactly.

Cost-Imposing vs. Advisory Local Measures (Full)

	Mean pro-renewable share (N)		Difference	Permutation	Pro side won, % (N)	
	Cost-imposing	Advisory	[95% CI]	<i>p</i>	Cost-imposing	Advisory
<i>A. Differences in means</i>						
All places	56.6 (41)	68.7 (47)	-12.1 [-19.1, -4.8]	0.003	70 (53)	77 (81)
Democratic share ≥ 60%	58.6 (36)	76.6 (37)	-18.0 [-23.0, -13.1]	0.000	75 (44)	92 (53)
Democratic share ≥ 70%	58.5 (30)	80.4 (23)	-21.9 [-27.1, -16.7]	0.000	71 (34)	97 (39)
Within the frame (F1, F2, F2b)	52.2 (21)	75.7 (39)	-23.5 [-30.4, -16.6]	0.000	61 (28)	84 (73)
Core tier (b): drops minor-share bundles	57.8 (35)	68.7 (47)	-10.8 [-18.3, -3.2]	0.010	76 (45)	77 (81)
Core tier (c): explicit, primary purpose	58.4 (21)	68.7 (47)	-10.3 [-19.3, -1.6]	0.051	73 (22)	77 (81)
Tier (b), Democratic share ≥ 60%	59.3 (33)	76.6 (37)	-17.4 [-22.8, -12.0]	0.000	80 (40)	92 (53)
Advisory excluding local siting votes	56.6 (41)	76.3 (38)	-19.7 [-25.1, -14.1]	0.000	70 (53)	87 (62)
and Democratic share ≥ 60%	58.6 (36)	77.4 (36)	-18.8 [-23.8, -13.7]	0.000	75 (44)	94 (51)
<i>B. Regression of pro-renewable share on cost-imposing indicator: coefficient (robust SE)</i>						
Bivariate	-12.1 (3.8)	N = 88				
Jurisdiction type and year	9.6 (8.4)	N = 88				
Jurisdiction fixed effects	-32.3 (6.8)	N = 13; 2 jurisdictions (Berkeley, San Francisco)				
Dem. share ≥ 60%, with Dem. share and year	-18.9 (2.6)	N = 73				
No siting advisories, jurisdiction type and year	-16.5 (6.7)	N = 79				

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Local Measures: Descriptives and Frame

	Measures	With returns	Pro side won, % (N)		Mean pro share	Median		
<i>A. By cost class</i>								
Cost-imposing (core set)	53	41	70 (53)		56.6	58.0		
Direct cost (tax, fee, bond)	38	28	84 (38)		60.5	61.0		
Compliance mandate	5	4	40 (5)		44.0	49.3		
Public ownership	10	9	30 (10)		49.9	48.3		
Advisory (non-binding)	85	47	77 (81)		68.7	76.7		
Binding, no new cost	16	9	86 (14)		63.1	57.6		
	Cost-imposing (core)				Advisory (no cost)			
	N	Returns	Won %	Mean	N	Returns	Won %	Mean
<i>B. By frame part</i>								
F1: Illinois census (1995–2026)	1	0	100	–	39	1	74	42.7
F1: California census (1996–2025)	13	13	31	51.6	3	3	100	82.4
F1: Massachusetts district questions	0	0	–	–	35	35	91	76.1
F2: Ballotpedia coverage units	9	5	78	49.2	0	0	–	–
F2b: same units, pre-coverage years	5	3	100	60.0	0	0	–	–
Outside the frame	25	20	80	61.1	8	8	12	34.5
<i>C. By period</i>								
2000–2009	15	13	60	54.0	20	11	90	81.5
2010–2017	14	10	79	58.6	24	9	50	35.8
2018–2026	24	18	71	57.3	41	27	86	74.4

Local Measures: Precinct Benchmarks

Measure	Cost	Precincts	Pro %	Dem %	Slope on Dem % (SE)	Pro — Dem, Dem ≥ 70% (N)	Same-ballot tax Yes %	Pro — tax
<i>A. Cases 2018–2026</i>								
Portland Measure 26-201 2018	direct	81	65.3	81.5	0.72 (0.04)	-16.7 (52)	69.0	-3.7
San Francisco Proposition A 2018	direct	597	77.2	90.9	1.34 (0.03)	-13.8 (597)	55.5	21.7
Berkeley Measure HH 2020	direct	33	47.0	95.9	0.28 (0.74)	-47.2 (33)	—	—
Denver Ballot Measure 2A 2020	direct	356	62.3	81.4	0.96 (0.03)	-19.7 (325)	—	—
Portland Question C 2020	mandate	11	49.8	67.2	0.81 (0.14)	-20.0 (5)	—	—
Ann Arbor Proposal 1 2022	direct	53	71.1	90.3	2.16 (0.29)	-16.7 (53)	63.1	7.9
Berkeley Measure I 2022	direct	31	59.4	94.7	-1.28 (1.20)	-31.0 (30)	64.9	-5.5
Boulder Ballot Issue 2A 2022	direct	58	70.5	90.7	0.72 (0.32)	-20.2 (58)	73.3	-2.9
Boulder Ballot Issue 2B 2022	direct	58	68.4	90.7	0.70 (0.32)	-22.2 (58)	73.6	-5.2
El Paso Proposition C 2022	direct	172	50.6	64.7	0.58 (0.02)	-18.2 (74)	57.1	-6.5
Burlington Question 2 2023	direct	8	67.3	—	—	—	—	—
Berkeley Measure GG 2024	direct	32	31.0	94.6	-2.84 (0.51)	-59.4 (30)	66.9	-35.9
Ipswich Question 6 2024	direct	4	44.3	65.5	—	-22.7 (1)	—	—
Portland Measure 26-253 2024	mandate	79	49.1	87.3	0.44 (0.04)	-39.1 (64)	—	—
Salt Lake City School District bond 2024	direct	140	61.1	78.3	0.39 (0.05)	-19.4 (110)	73.6	-12.5
<i>B. Earlier cases, 2001–2017 (supporting)</i>								
San Francisco Proposition B 2001	direct	648	72.6	82.0	0.91 (0.04)	-10.1 (477)	72.1	0.4
San Francisco Proposition F 2001	ownership	648	49.8	82.0	1.30 (0.04)	-31.0 (477)	72.1	-22.3
San Francisco Proposition H 2001	direct	648	53.9	82.0	1.08 (0.05)	-28.5 (477)	72.1	-18.2
San Francisco Proposition I 2001	ownership	648	48.3	82.0	1.28 (0.04)	-32.5 (477)	72.1	-23.9
San Francisco Proposition D 2002	ownership	631	46.3	81.6	1.24 (0.03)	-35.1 (550)	56.2	-9.9
Boulder County Ballot Issue 1A 2008	direct	234	63.7	72.9	0.83 (0.02)	-10.8 (142)	67.4	-3.7
San Francisco Proposition H 2008	ownership	579	38.6	86.0	1.45 (0.04)	-47.4 (570)	73.7	-35.0
Boulder Issue 2B 2011	direct	88	50.4	85.1	1.57 (0.11)	-33.2 (87)	74.9	-24.5
Boulder Question 2C 2011	ownership	88	51.9	85.1	1.58 (0.13)	-30.8 (87)	62.5	-10.6
Boulder Ballot Issue 2A 2012	direct	86	82.3	83.5	0.34 (0.11)	-1.5 (84)	85.0	-2.7
Boulder Question 2E 2013	ownership	87	66.5	86.2	1.17 (0.06)	-19.6 (86)	65.8	0.7
Boulder Question 310 2013	ownership	87	68.7	86.2	1.25 (0.08)	-16.4 (86)	65.8	2.8
Lafayette Issue 301 2013	direct	17	50.4	71.1	0.75 (0.09)	-22.0 (11)	17.4	33.0
Lafayette Ballot Issue No. 2D 2014	direct	17	45.8	71.3	0.56 (0.11)	-27.4 (11)	60.7	-14.9
Boulder Ballot Question 2P 2015	direct	87	77.2	86.2	0.98 (0.06)	-7.5 (86)	57.6	19.6
Boulder Issue 2L 2017	direct	87	51.7	86.2	1.23 (0.15)	-31.3 (86)	80.0	-28.3
<i>C. Pooled, measure fixed effects (SE clustered by measure)</i>								
2018–2026 cases		1702			0.72 (0.11)	-19.6 (2.8)		
All cases, 2001–2026		5944			0.93 (0.10)	-26.2 (3.4)		

D. Ballot-level benchmark, 2020 cast vote records

	This measure			Other tax measures, same ballot		
	Dem. Yes	Rep. Yes	APG	Dem. Yes	Rep. Yes	N
Denver 2A (climate sales tax)	75.5	12.4	63	81.8	29.1	3
Berkeley HH (utility tax)	49.4	5.3	44	75.9	23.4	5
Columbus Issue 1 (no-cost aggregation)	89.7	38.8	51	69.5	43.8	5
National local tax/spend benchmark, 2020	71.8	52.4	20			

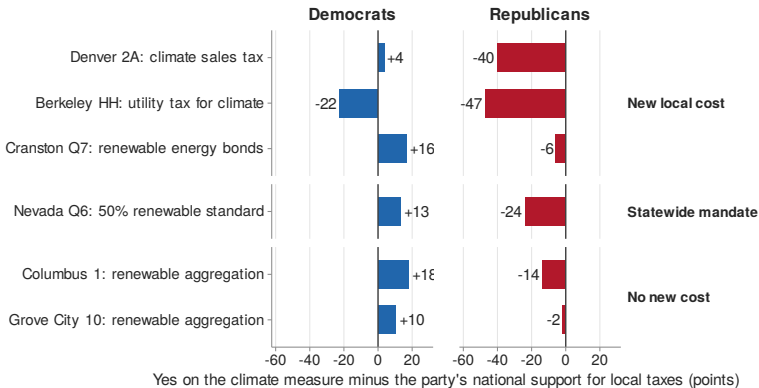
Local Measures: How the Frame Was Built

- ◉ **F1:** official lists of every local question: Illinois (1995–2026), California Elections Data Archive (1996–2025), Massachusetts
- ◉ **F2:** Ballotpedia's complete coverage: 100 largest cities from 2018, state capitals from 2021, all of California from 2016, NC 2019, VA 2025–26
- ◉ **F2b:** the same cities back to 2000 in official archives, costly measures only
- ◉ Screens: 12,149 IL referenda; 6,243 Ballotpedia measures; 11,501 CA measures; audits of 250 rejects bound misses at 1.2–2.2%
- ◉ Topic tags alone would miss half the costly measures (8 of 16)
- ◉ Cost coded from official (33 of 53) or secondary (14) ballot text, before outcomes
- ◉ Tiers: (b) drops minor-share bundles; (c) explicit, primary climate purpose

Cast Vote Records, 2020 (Full)

	Yes vote (%)			Party gap (Dem. – Rep.)	vs. party's national local tax support		vs. same voters' other local taxes on ballot	
	Dem.	Rep.	Split		Dem.	Rep.	Dem.	Rep.
<i>A. Local measures that impose a new cost</i>								
Denver, CO, Measure 2A (climate sales tax)	75.5	12.4	40.9	63.1	+3.7	-40.0	-6.1	-16.4
Berkeley, CA, Measure HH (utility users tax for climate)	49.4	5.3	16.2	44.1	-22.4	-47.1	-25.8	-17.7
Cranston, RI, Question 7 (\$5M renewable energy bonds)	88.3	46.5	66.6	41.8	+16.5	-5.9	+4.5	-19.2
<i>B. Statewide renewable mandate</i>								
Nevada Question 6 (50% renewable portfolio standard)	85.0	28.6	63.7	56.4	+13.2	-23.8	–	–
<i>C. Local measures with no new cost</i>								
Columbus, OH, Issue 1 (renewable electricity aggregation)	89.7	38.8	73.2	50.9	+17.9	-13.6	+5.4	-5.6
Grove City, OH, Issue 10 (renewable electricity aggregation)	82.2	50.2	77.0	32.0	+10.4	-2.2	+7.1	-0.3

2020 Cast Vote Records vs. the Party's National Support for Local Taxes



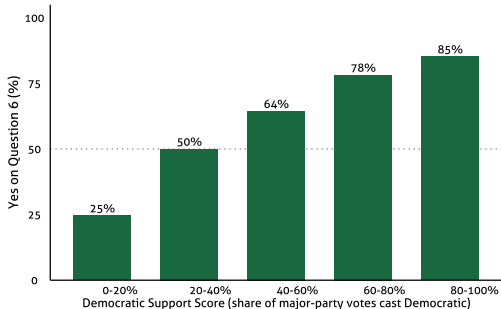
Yes vote on each measure minus the party's national 2020 support for local tax and spending measures (Dem. 71.8%, Rep. 52.4%).

California Cast Vote Records (Full)

County, year	Climate measure (paired measure)	Yes vote (%)			Party gap (Dem. — Rep.)	vs. same voters' vote on paired measure			Ballots
		Dem.	Rep.	Split		Dem.	Rep.	Split	
Marin, 2024	Prop. 4, climate bond (\$10B) (Prop. 2, school bond (\$10B))	82.4	23.0	54.0	59.4	+7.5	-10.5	+6.3	139,189
Santa Barbara, 2024	Prop. 4, climate bond (\$10B) (Prop. 2, school bond (\$10B))	84.2	20.6	55.1	63.6	+4.7	-6.0	+3.9	187,876
Marin, 2022	Prop. 30, income tax for zero-emission vehicles (Prop. 28, arts and music funding)	57.1	13.0	32.9	44.1	-27.4	-16.1	-24.0	114,402

Notes: County cast vote records. Party of US House vote; split-ticket: House and top-of-ticket votes (Governor 2022, President 2024) differ; the top of the ticket classifies voters without a House vote. Comparison: each voter's climate vote minus the paired measure on the same ballot, averaged within party (points). Marin and Santa Barbara are the counties whose records link every card; both lean Democratic.

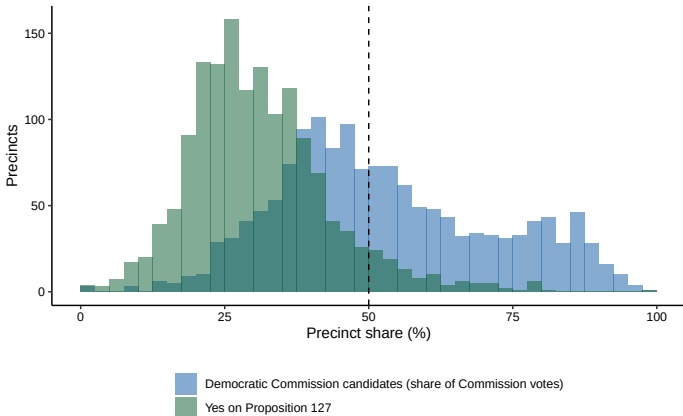
Nevada Question 6 by Democratic Support Score



Yes share on Question 6 (2020) by the ballot's Democratic Support Score: share of its partisan contests won by the Democrat.

- 85% among straight-ticket Democrats, 24% among straight-ticket Republicans
- Passed twice (2018, 2020) under conditions unique to Nevada: a credible economic case from solar resources, and no serious organized opposition

Arizona 2018: Both Distributions



Precinct support for the Democratic Commission candidates and for Proposition 127, Arizona 2018.

C. Interest Groups

Measuring Effort Inside the Beltway

Each docket record: organization and family, reach, proceeding type, document type, pages, date; each request records issue, action, position, and disposition in the final order

Preparation-cost equivalent of document d of type k :

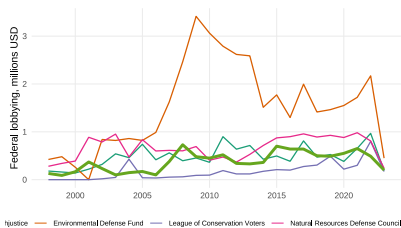
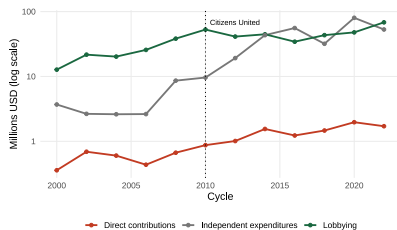
$$C_d = r_s(a_k + b_k P_d)$$

- ◉ P_d prepared pages; a_k fixed preparation time; b_k hours per page
- ◉ r_s : \$300/hour (2023 dollars, from intervenor-compensation materials), or scaled by the state's lawyer wage (BLS OEWS)
- ◉ Benchmarked against organizational budgets (IRS Form 990)

Filers: green NGOs, renewable firms, utilities, fossil fuel firms, other non-profits; public advocates separate; counsel is not the client; joint filings attributed once

Federal Lobbying and Campaign Finance, 2000–2022

Coalition	Lobbying dollars per dollar of	
	direct contributions	direct contributions and independent expenditures
Environmental organizations	35.9	1.4
Fossil fuel producers and utilities	23.5	23.4



Environmental-sector federal spending by channel (log scale)

Federal lobbying by focal NGOs; Sierra Club in bold

With independent expenditures, greens match industry after 2014: episodic mega-donor spending in a few salient federal races, which does not scale down to a Commission race. OpenSecrets JE300; industry codes exclude nuclear, renewables, water

Georgia PSC: Candidate Content

	Candidacies
Total candidacies coded	65
No identifiable position	30
Lean pro-renewable	18
Strongly pro-renewable	9
Mixed or conflicting	4
Lean or strongly anti-renewable	4
League of Conservation Voters endorsement	0
Sierra Club endorsement	3

Notes: Hand coded Georgia PSC candidacies from 1998–2022 based on campaign coverage, debate records, ballot guides, and endorsement archives; each coding carries a documented evidence summary and sources. Categories are mutually exclusive.

2024 reviews of races in all eight electing states: platforms organized around reliability, affordability, and procedure; renewable energy largely absent

Allocation Across Channels: Tables

Resource allocation across channels

Sample	Window	Inside / direct	Inside / comparison
Federal: Green	2000–2022 cycles	35.90	1.38
Federal: Incumbents	2000–2022 cycles	23.49	23.45
State: green	2017–2024	—	38.42
State: incumbent	2017–2024	—	2.34

Organizational resources by channel, per organization–state–year

Coalition	Campaign \$	Campaign budget %	Lobbying \$	Lobbying budget %
Green	4,045	0.105	155,388	4.051
Incumbent	104,537	2.165	244,180	5.065
Adjusted green contrast	-3.268 (0.019)	-0.021 (0.000)	-0.506 (0.058)	-0.010 (0.002)

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Selection Design and Time Allocation

Relative resources and selection design

System / contrast	Channel	Dollar ratio / log contrast	Budget ratio / fraction contrast
Appointed	lobbying	0.737	0.929
Appointed	campaign	0.038	0.049
Elected	lobbying	0.283	0.349
Elected	campaign	0.040	0.048
Green $\times$ elected	lobbying	-0.947 (0.032)	-0.030 (0.002)
Green $\times$ elected	campaign	0.056 (0.044)	0.000 (0.001)

Time allocation among NGO participants (189 organization–state–years)

Activity	Mean hours	Org-state-years
Docket preparation	641.5	189
Other inside activity	55.4	189
Total inside activity	696.9	189
Campaign labor	20.0	189
Inside / campaign mean-hours ratio	34.91	

Docket Samples and Participation

Matched pairs

Pair	Elected	Appointed	Region
1	AZ	NV	West
2	GA	KY	South
3	LA	AR	South
4	MS	WV	South

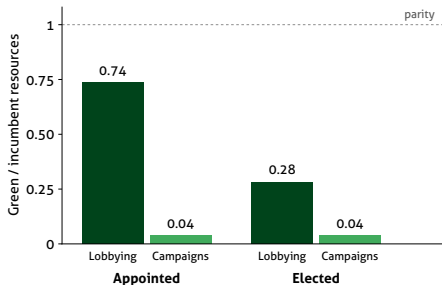
Participation and filings, first 180 days

Actor class	Participation	Filings	Substantive	Priced
Consumer nonprofits	36.9%	918	61.1%	92.9%
Fossil-fuel firms	27.5%	675	62.8%	93.6%
Environmental NGOs	91.6%	5,210	69.3%	92.8%
Public advocates	74.1%	1,778	62.2%	93.3%
Renewable firms	28.9%	710	61.3%	92.8%
Utilities	97.4%	3,548	61.6%	93.1%

Organizational Practices and Competitiveness

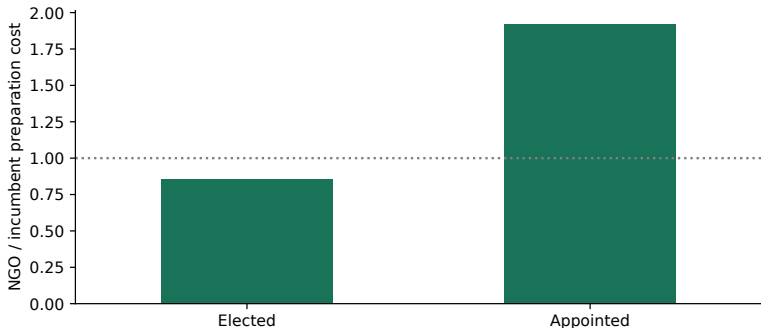
Filings in appointed systems,
first 180 days

Actor	Substantive	Testimony	Joint
Fossil-fuel firms	63.5%	21.9%	0.0%
Environmental NGOs	73.4%	27.2%	17.4%
Utilities	62.3%	22.3%	0.0%



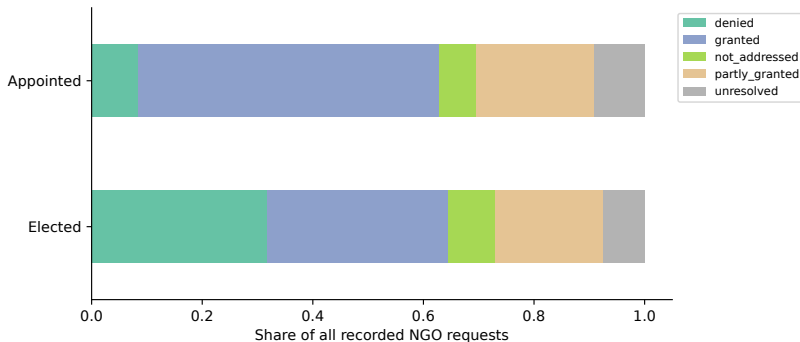
Green-to-incumbent ratios of mean resources;
one = equal measured resources (not an
equivalence test)

Preparation Resources Relative to Incumbents



NGO preparation resources relative to incumbent industries within eligible docket cohorts. The reference line marks equal aggregate measured resources, not equal access or influence.

Disposition of All Recorded NGO Requests



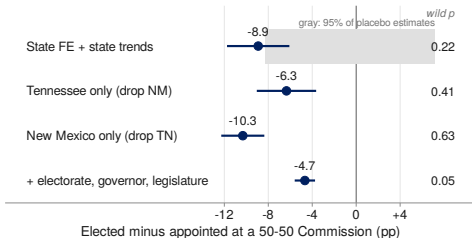
Contrast	Estimate	95% interval
Appointed minus elected	0.333	[0.266, 0.399]

Coding Requests and Final Orders

- ⦿ One row per organization, docket, and distinct requested action, with page or paragraph location in a dated filing
- ⦿ Disposition located in the final order: granted, partly granted, denied, not addressed, unresolved, withdrawn, unclear
- ⦿ Partial grants kept separate; unresolved and unaddressed requests are *not* defeats
- ⦿ Requests averaged within organization–docket, so long briefs do not add independent cases
- ⦿ The order date sets the regime; transition-spanning proceedings flagged; effort must precede the order
- ⦿ Joint filings attributed once (lead author or largest documented contribution; equal shares as sensitivity)
- ⦿ Second coder on a stratified subset

D. Anticipated Questions

Two Switching States: Strength and Limits



- **Why within-state:** elected states are windier, more rural, more Republican, so comparing across states mixes selection with geography
- **What the two cases add:** the same state under both systems; the switches run in opposite directions (TN to appointment; NM to election and back); each state alone gives a negative gap (-6.3 , -10.3) once its own trajectory is modeled
- **Where it stops:** two clusters, so wild-bootstrap $p = 0.05$ – 0.63 ; the estimate sits just outside 95% of placebo estimates (placebo $p = 0.009$)
- $\Rightarrow$ Read as two careful case studies, and test the mechanisms directly (Parts II, III)

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Discounting and the Social Cost of Carbon

- ◉ **Discount-rate debate:** the weight given to future damages drives the social cost of carbon; a low rate implies aggressive action now (Stern 2007), market-based rates a gradual ramp (Nordhaus 2007; Weitzman 2007)
- ◉ **Present bias:** quasi-hyperbolic discounting shrinks delayed benefits and makes choices time-inconsistent (Laibson 1997); citizens discount policy benefits that arrive later (Jacobs and Matthews 2012)
- ◉ **Energy-efficiency gap:** consumers pass up efficiency investments that pay for themselves, at high implicit discount rates (Hausman 1979; Jaffe and Stavins 1994; Allcott and Greenstone 2012)
- ◉ **Politics of the long term:** voters reward disaster relief, not preparedness (Healy and Malhotra 2009); long-term investment must overcome electoral short-termism (Jacobs 2011)

→ The gap between voters and the welfare benchmark is the same question: whose preferences count, over what horizon

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Aren't Voters Just Discounting the Future Rationally?

- ◉ Possibly, and I take revealed preferences on ballot measures as sincere
- ◉ My benchmark is social welfare, which includes the externality and future consumers; voters' own choices need not include either
- ◉ The gap may be a genuine distributional conflict between today's energy consumers and future generations, the same question as the discount-rate debate (Stern 2007; Nordhaus 2007)
- ◉ The empirical claim does not depend on resolving it: whatever its source, the electoral signal elected Commissioners face runs against acting on the transition

Why Prefer the Trend-Inclusive Estimate?

- ⦿ State FE absorb each state's time-invariant endowment; the remaining difference between specifications is the *trajectory* of development
- ⦿ Without trends, the Plains wind build-out (production tax credit, resource quality) coincides with the elected states and is attributed to the institution
- ⦿ With trends, the difference is identified from deviations around each state's own path
- ⦿ Trade-off: trends also absorb an institutional effect that works *gradually*, through the slow accumulation of resource-plan and rate-case decisions, so they may understate it

Isn't the Appointer Rule Mechanical?

- ◉ It is partly tied to the governor by construction, so every composition result is also shown with each Commissioner's own party (MSU archive)
- ◉ On personal party, appointed Commissions still follow the governor: a Democratic governor is associated with a Commission 38 pts more Democratic (61 on the appointer rule)
- ◉ Mismatch cases: appointed Commissions in blue states with Republican governors are as Republican as those in red states (31% vs. 32% Democratic)
- ◉ Disagreement with personal party (72% agreement for appointed seats) is plausibly closer to classical error → attenuation
- ◉ Where the two measures disagree (the blue-state gradient), I say so: the Democratic-appointer gradient is not established

Is It Market Structure or Generation Authority?

- ◉ Every elected Commission sits in a vertically integrated state; many appointed ones are restructured
- ◉ The mechanism requires the gap where the Commission governs generation: IRPs, certificates, rate-basing
- ◉ Elected–appointed difference at 50–50, state FE and trends:
 - ◉ controlling for retail choice: -8.8 pp
 - ◉ never-restructured states only (35): -8.7
 - ◉ excluding TN (TVA) and NE (public power): -10.4
 - ◉ both restrictions: -9.8
- ◉ Same caveat as the main estimate: identified from the switching states

► Table

Are Ballot Measures a Selected Sample?

- ◉ Yes, and the selection likely *inflates* support: measures need organized backing to qualify, and initiative states are disproportionately Western and track majority opinion closely (Matsusaka 2004; Lupia and Matsusaka 2004)
- ◉ Statewide list is complete for 2000–2024; local completeness is claimed only within a frame fixed before looking at outcomes
- ◉ Costly and advisory sets are defined by question content, not outcome: a costly measure that passed counts as much as one that failed
- ◉ Places differ: the cleanest comparison is within Berkeley and San Francisco (–32 pts) and against same-ballot taxes
- ◉ CVR counties lean Democratic, so they describe those electorates, not the state

Does Federal Behavior Tell Us About the States?

- ◉ No comprehensive disclosure of state-level lobbying exists; the federal record is the one place both channels are fully measured for the same actors
- ◉ The organizations are the same at both levels: Sierra Club, NRDC, EDF, LCV operate in Washington and in state capitals
- ◉ The allocation reveals the technology of influence they have built, which travels with the organization
- ◉ The state evidence (Commissioner races, ballot committees, dockets, NM Amendment 1) then asks whether they behave the same way where the stakes are Commissioners
- ◉ Small campaign contributions are not evidence of small influence (Ansolabehere et al. 2003), which is why I measure inside effort directly

Isn't New Mexico a Single, Bundled Case?

- ◉ Yes: the 2023 change bundled selection method, membership, and size ($5 \rightarrow 3$), alongside other policy and economic developments
- ◉ Amendment 1 is revealed-preference evidence, not an estimate of an effect: the same organizations that decline to fund Commissioner candidates paid to end Commissioner elections
- ◉ In the docket analysis: 2020–2022 anticipation kept separate; transition-spanning proceedings flagged and excluded as a sensitivity check; utilities as a comparison series, not a placebo
- ◉ Tennessee also switched (1996), but its proceeding records before 2000 could not be obtained

Is Appointment Always Better for the Transition?

- ⊙ No. Appointed Commissions follow the governor, so a Republican governor in a blue state produces a Commission as Republican as a red state's
- ⊙ Appointment is where green interests have their best *chance* to compete, not a guarantee
- ⊙ Expectation not tested in the main talk: the elected–appointed gap should be largest when the selecting principal is Democratic (the governor's coalition includes green interests and renewable developers). The pooled margin plots in Appendix F are suggestive but cross-state

Why Would Pro-Consumer Groups Prefer Less Democracy?

- ◉ Not less democracy in general: an arena where their resources (expertise, testimony, persistence) count, and where incumbents' electoral advantages (money, votes, a simple message) sit idle
- ◉ Under appointment the governor builds the coalition for them; they need not assemble an electoral majority
- ◉ Consistent with Fournaies and Fowler (2022) on insurance regulation: influence runs largely through channels other than transfers to officeholders
- ◉ Normative question left open: rational adaptation to US institutions, or a departure from a grassroots mandate? What does it mean for the legitimacy of appointed regulation?

E. The Argument in Detail

The Argument in Brief

Within this new interest group structure, **electoral accountability gives rise to regulators who are less likely to promote a long-term energy transition**, and more likely to be staunchly pro-incumbent industry

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- ◉ **Insulated** regulators can maximize across the information and support they receive from organized interests

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- Green interests are overshadowed in elections: incumbents' decades of presence, status-quo bias, a simpler message; few voters pay more now for renewables
- More important: green interests are **strong inside the beltway**, and have adapted to prioritize lobbying over campaigns

→ Lower renewable growth on average where voters elect the Commission, varying with the electorate's partisanship, preference distributions, and resources

How Regulator Selection Affects Alternative Energy: The Electoral Arena

Elections are themselves arenas in which organized interests can dominate diffuse voters (Stigler 1971). The pro-transition coalition is weak on both flanks

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- ◉ **Business flank outsized:** **alternative energy firms** vs. **fossil fuel producers**, among the most politically organized industries, and **monopoly utilities**, who campaign at a scale entrants cannot match
 - ◉ Incumbents also command votes, through the workforces and communities that depend on them (Bombardini and Trebbi 2020)
- ◉ **Consumer flank small:** the largest bloc is status-quo consumers who care about price, and their message aligns with the more powerful business interest
 - ◉ Fossil fuel firms' long-lived political apparatus cemented economic identities that leave consumers skeptical of renewables (Dewitte 2026)
 - ◉ Voters willing to pay more for renewables are a small subset, even in the most Democratic states

The Electoral Arena: A Hard Message to Carry

- Benefits of the transition are complex: externalities and short-versus long-term trade-offs, with no ready heuristics (Lupia 1994)
- Infrastructure adds visible short-term costs, and projects disturb communities (offshore wind at Martha's Vineyard; siting backlash: Stokes 2016)
- 'Lower prices now' is simple, backed by the dominant business interest, reinforced by status-quo bias
- Elected commissioners act as if punished for price increases (Parinandi and Hitt 2018)
 - AZ Democrats won on renewables and rates in 2018, lost 2022 and 2024; GA's 2025 Democratic PSC wins ran on rates and Vogtle; Davante Lewis (LA 2022) the exception
- A natural majority that never is one at the ballot box (Przeworski and Sprague 1986)
- Off-cycle, down-ballot races favor narrow interests (Anzia 2011); initiatives return groups to an electorate with modest willingness to pay (Ansolabehere and Konisky 2014)

Regulator Selection Under Industrial Change: Inside the Beltway

Electoral accountability tilts regulator behavior toward status quo (fossil + utility interests) → slower integration of alternative energy

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Key difference: alternative energy firms and pro-alternative energy NGOs have more influence **INSIDE THE BELTWAY**

- Influence depends on *relative* efficiency at producing pressure; arenas reward different technologies of pressure (Becker 1983)
- The task inside: show regulators integration is best for long-term pricing and stability
- Green groups compete with what they hold: expertise, testimony, persistence in resource-plan and rate cases, regulators' technocratic language
- The governor assembles the coalition for them: green voters bundled with monied allies (renewable developers, in-state manufacturers)

Inside the Beltway: Strategic Allocation

- ⊙ Appointed regulators maximize across organized interests' information and support; green NGOs punch above their electoral weight → **more balanced regulators**
- ⊙ Where fossil fuel interests dominate elections unopposed, regulators are more likely staunchly pro-fossil fuel
- ⊙ Green organizations should lobby, intervene, and cultivate appointers, and spend on elections sparingly and episodically
- ⊙ It is the *return to inside effort* under each institution; inside activity need not center on appointments
- ⊙ Consistent with insurance regulation: influence runs largely through channels other than transfers to officeholders (Fouirnaies and Fowler 2022)

Measurement implication: small campaign contributions are not evidence of small influence (Ansolabehere et al. 2003) → measure effort inside the regulatory process directly

F. Moved from the Main Talk

Empirical Expectations

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 - ⊙ For a given effort, their requests fare better before appointed Commissions
3. **Voters:** support for renewable energy is low, even among Democrats, when it carries a direct, short-term cost

Data and Sources

- ◉ **Outcome:** share of generation from wind, solar, geothermal, and biomass; 50 states, 1990–2024 (EIA)
 - ◉ Excludes hydro: a legacy of dam building, fixed by geography, that predates every Commissioner [▶ With hydro](#)
- ◉ **Commissions:** selection method from constitutions and statutes; original roster of all 1,287 Commissioner spells, 1990–2026
 - ◉ Appointed Commissioners have no official party: I use the appointing governor's or legislature's party; 97% (elected) and 72% (appointed) agreement with personal party where known [▶ Measure](#)
- ◉ **Electorate:** Snyder (2026) normal vote across all statewide contests; governor's party; legislative seats
- ◉ **Controls:** log GDP, manufacturing share, log population (lagged); NREL wind and solar endowment; RPS and restructuring ledgers

Energy on the Ballot

Measure type	Statewide	Local	Total
Fossil fuels, facility siting, and other energy	13	204	217
Bond, tax, and finance	12	70	82
Renewable mandates and subsidies	11	64	75
Utility governance	7	0	7
Public power, transmission, and aggregation	5	1,056	1,061
Carbon pricing and climate policy	5	23	28
Nuclear	0	5	5
All energy measures	53	1,422	1,475

Green Groups Give About 1% of Campaign Money

One high-profile energy referendum vs. 25 years of Commissioner elections in the archetypal elected state

Context	Total spending	Green nonprofit \$
Maine Q1 Yes side (Block NECEC)	\$27.4M	\$284k
Maine Q1 No side (Allow NECEC)	\$66.7M	\$0
<i>ME Q1 Total</i>	<i>\$94.1M</i>	<i>\$284k (0.30%)</i>
GA PSC Democratic candidates, 1998–2022	\$4.3M	\$49k
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<i>GA PSC Total</i>	<i>\$9.9M</i>	<i>\$110k (1.1%)</i>

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- Sierra Club Georgia: \$3,500 to PSC candidates in 25 years; energy-sector money 91% to Republican incumbents, solar firms included

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Green Money Not Lacking Because No Green Candidates

Position on renewables	N	Raised (\$M)	Pro-env. (\$k)	Alt.-energy firms (\$k)	Green share (%)
Strongly pro-renewable	9	2.68	10.9	4.6	0.6
Lean pro-renewable	18	2.12	7.0	22.4	1.4
Mixed or conflicting	4	0.62	5.0	24.0	4.7
No identifiable position	30	2.85	9.9	6.2	0.6
Lean or strongly anti	4	1.63	6.6	8.0	0.9
All candidacies	65	9.90	39.5	65.2	1.1

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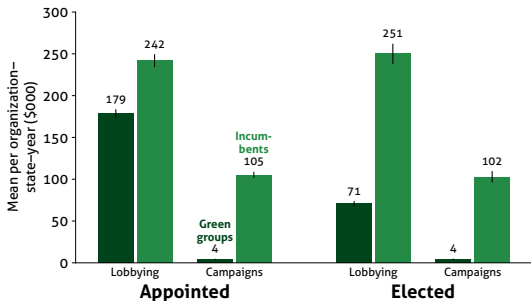
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- ◉ Nearly half take no position; zero LCV endorsements, three Sierra Club
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- ◉ Sierra Club-endorsed Lindy Miller raised \$1.7M (2018), \$201 of it from pro-environmental donors

Green Lobbying Rivals Incumbents' Only Under Appointment

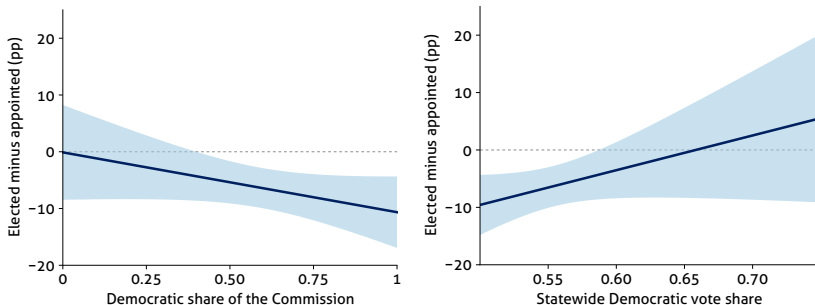


Resource allocation 2017–2024, per organization–state–year. Intervals resample states.

Green / incumbent mean resources	Lobbying	Campaign
Appointed systems	0.74	0.04
Elected systems	0.28	0.04

Under appointment, green lobbying approaches incumbents'; campaign resources are a small fraction in both [► Tables](#)

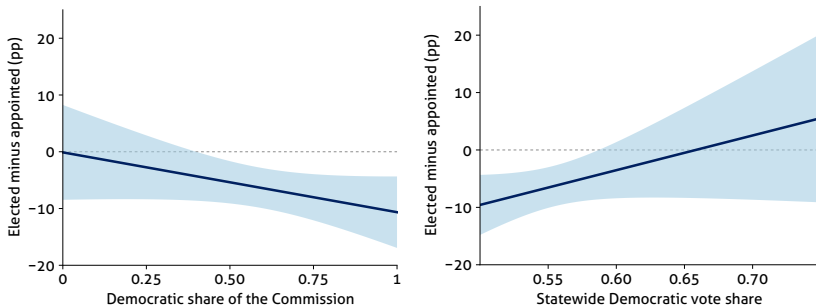
Is the Gap Concentrated Among Democratic Commissions?



Marginal effect of an elected Commission, Democratic vote share ≥ 0.5 . Left: by composition (personal party). Right: by electorate's Democratic share.

- Personal party (left): the gap widens as the Commission becomes more Democratic

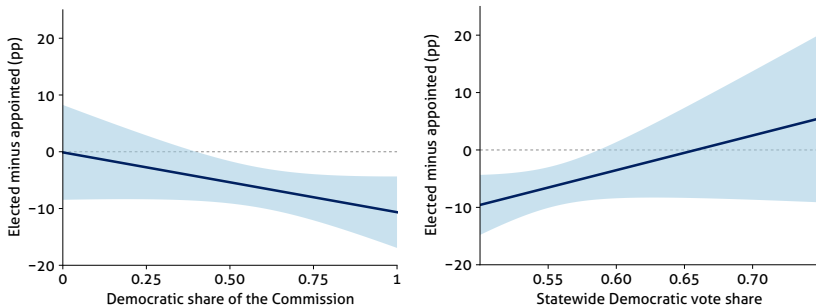
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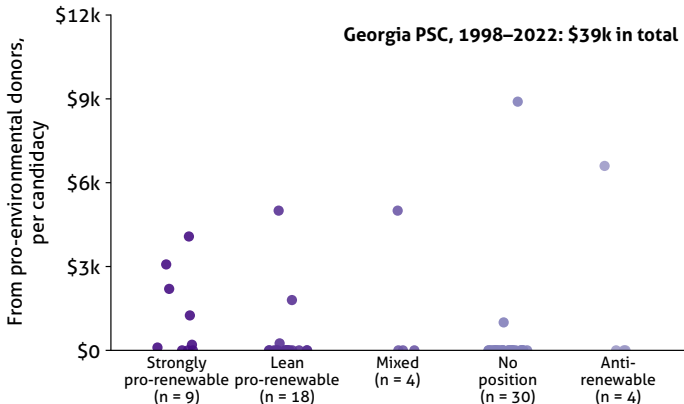
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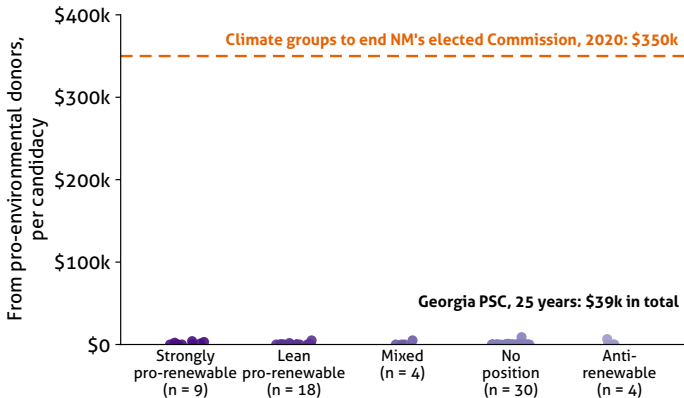
Georgia, 1998–2022



65 Georgia PSC candidacies, 25 years: \$39k from pro-environmental donors (0.4% of \$9.9M). Strongly pro-renewable candidates got about the same share as those with no position

Follow the Money, 1998–2022; renewable positions hand-coded for every Georgia PSC candidacy.

Georgia, 1998–2022



Green groups barely fund Commissioner candidates, even pro-renewable ones, yet paid \$350k in one campaign to end Commissioner elections

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[◀ Back](#)

Why Regulators

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- ◉ With rapid industrial and technological change, regulator decisions are likely increasingly important
- ◉ Theory and empirical evidence indicate that regulator behavior can determine important industry outcomes, which varies based on how they are selected
- ◉ In the US, regulators are either *appointed* (most common) or *elected* directly by voters (a non-trivial number of states)

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- ◉ Evidence this shapes industry outcomes (Besley and Coate 2003, Leaver 2009, Lim and Yurukoglu 2018)
- ◉ The standard view: direct election → more consumer-oriented policy

New Interest Group Preference Structure

Consumers

- ◉ **Pro-Green consumers:** Consumer blocks who prefer industry change over lowest possible prices due to associated externalities
In theory, this group could be important for electoral outcomes. In practice, voters explicitly willing to pay more for renewables are a small subset, even in the most Democratic parts of the US

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Membership-based and 'grassroots', but staffed by highly educated professionals with elite access (Skocpol 2003). Can act through campaigns, or inside the regulatory and appointment process

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Business Interests

- ◉ **Renewable Energy Producing Firms:** Substitutes for firms in the existing business interest group, against **fossil fuel producers and naturally monopolistic utilities** that have long dominated the regulatory arena

Why Elections May Underinvest in Climate Policy

Even with rational voters and a decisive median voter, elections likely fall short of max social welfare

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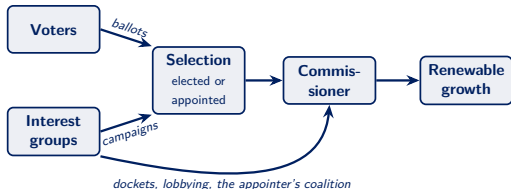
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→ Beyond energy: infrastructure, disaster preparedness (Jacobs 2011; Healy and Malhotra 2009)

In theory, voters underinvest. How much do they actually behave this way?

▶ Classic debates

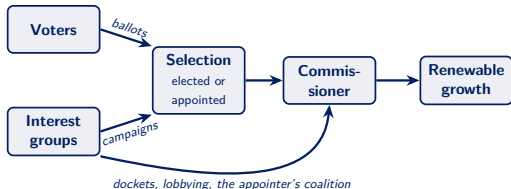
When Should Selection Matter Most?



Important Variation

- Partisanship of state government, and of the Governor (appointer) in particular

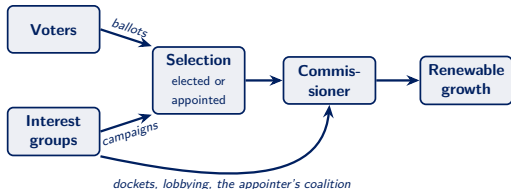
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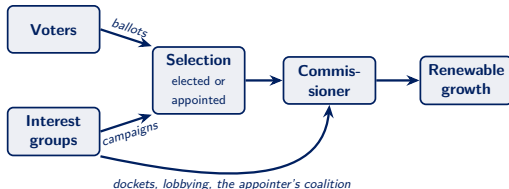
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- ◉ Underlying preference distributions
- ◉ Commission authority over generation: vertically integrated vs. restructured; TVA (TN) and public power (NE) outside PUC reach

What Happens When Industries Undergo Change?

Heavily regulated sectors oft experience large-scale industrial change

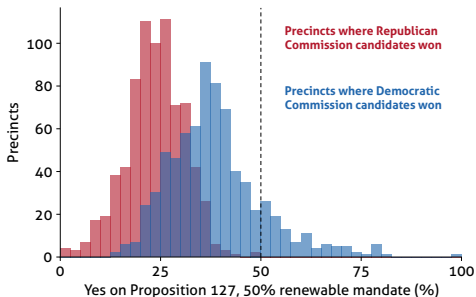
This induces preference heterogeneity *within* existing business and consumer groups

→ Creates new, cross-cutting coalitions

→ Regulators no longer face pro-business incentives when appointed and pro-consumer incentives when elected

Economics literature: railroads used truck-weight rules against a substitute technology (Stigler 1971); regulatory bargains erode when technology shifts the coalition beneath them (Peltzman 1976, 1989).

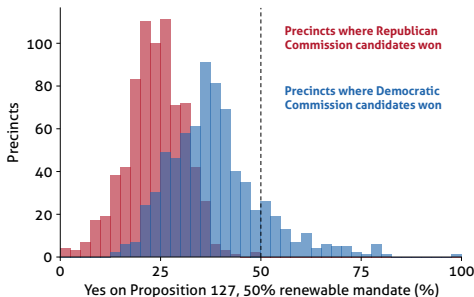
Arizona 2018: Candidate vs. Policy on the Same Ballot



Precinct support for Prop. 127 (50% RPS), by whether precinct was Dem Commission majority.

- Only election with Commissioners and a renewable mandate on one ballot

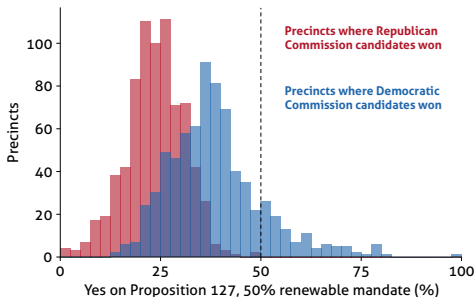
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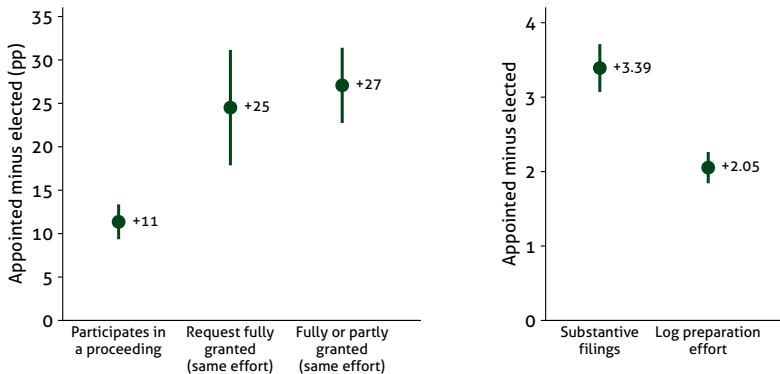
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- 85%** of Democratic-candidate-majority precincts voted **No**; where Yes won, the two Dems ran within ~2 pts: a party vote, not an issue vote

Participation, Effort, and Dispositions



Effort is per eligible proceeding (no green participation counts as zero): it combines participating more often with doing more once there [▶ Table](#)

Appointed minus elected, four matched pairs; 95% intervals (state-clustered CR1).

Matched Pairs: Table

Rows 1–3 per eligible proceeding (no green participation counts as zero);
rows 4–5 per decided request

Appointed minus elected	Estimate	95% interval
Participated in proceeding	0.114	[0.095, 0.132]
Substantive filings	3.39	[3.09, 3.69]
Log preparation cost	2.05	[1.86, 2.24]
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- Limits: 8 states; recorded work is not all work

Voters and Climate Policy: What We Know

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Revealed support: single measures, disasters, parties

- ◉ California environmental initiatives track income, ideology, partisanship (Kahn and Matsusaka 1997; Wu and Cutter 2011)
- ◉ Ideology explains most support for Washington's carbon tax (Anderson, Marinescu, and Shor 2023)
- ◉ Wildfires raise pro-environment voting among Democrats only (Hazlett and Mildenberger 2020); disasters and Swiss referendums (Baccini and Leemann 2021)
- ◉ Voters punish incumbents for local wind siting (Stokes 2016)
- ◉ Green-party support is different: a party vote imposes no direct cost, and support for a party varies and can also strategically motivation.

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Revealed support: single measures, disasters, parties

- California environmental initiatives track income, ideology, partisanship (Kahn and Matsusaka 1997; Wu and Cutter 2011)
- Ideology explains most support for Washington's carbon tax (Anderson, Marinescu, and Shor 2023)
- Wildfires raise pro-environment voting among Democrats only (Hazlett and Mildenberger 2020); disasters and Swiss referendums (Baccini and Leemann 2021)
- Voters punish incumbents for local wind siting (Stokes 2016)
- Green-party support is different: a party vote imposes no direct cost, and support for a party varies and can also strategically motivation.

Do voters, including Democrats, support renewable energy? Particularly *when it costs them*?

Full review: companion paper on the universe of energy and climate ballot measures.

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Behavioral Measure of Voter Support for Renewable Energy

Analytical Unit:

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Analytical Unit:

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- Individual ballots: Cast Vote Records (Baltz et al. 2024)